THREE-HINGED FRAMES

- Statically determinate systems which have three hinges are called as «**three-hinged frames**».
- In Figure 1, one of the hinges is on the system and the other two of them are on supports.
- In Figure 2, if the supports are fixed supports, 3 hinges can be on the system elsewehre from the supports.

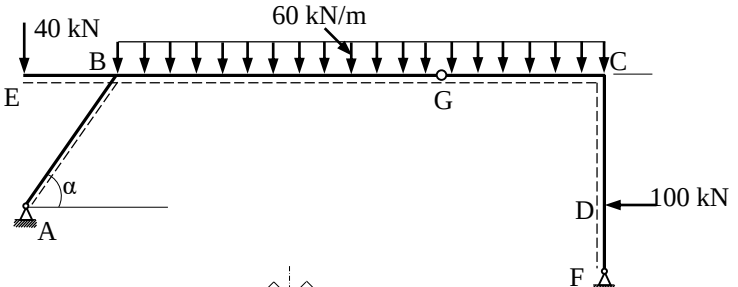


Figure 1

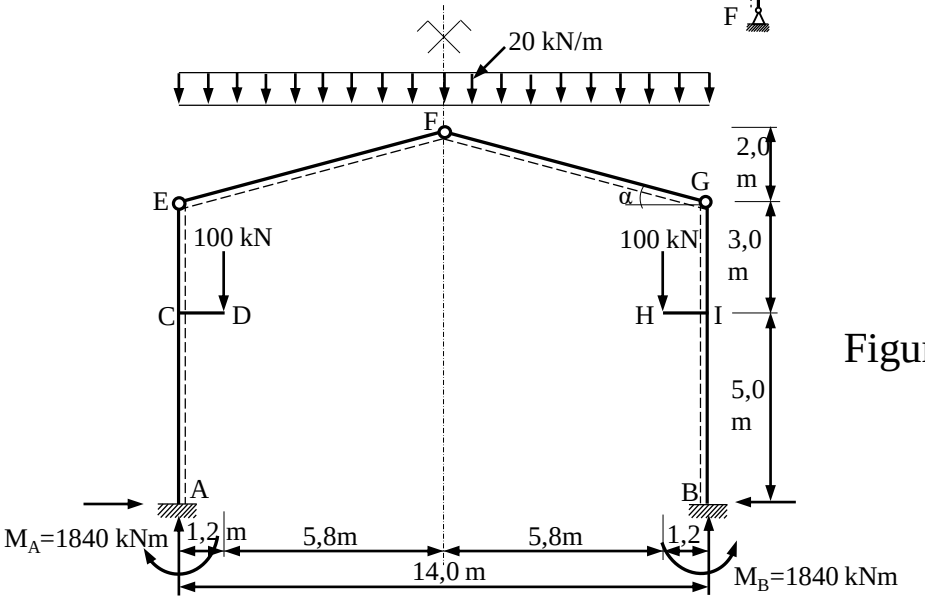
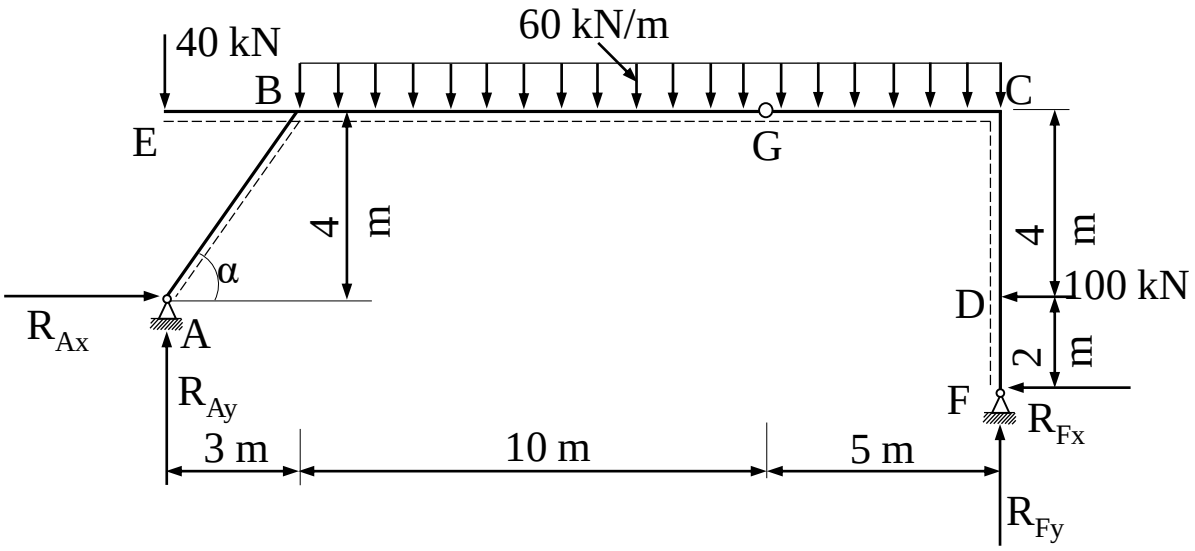


Figure 2

Example 1:



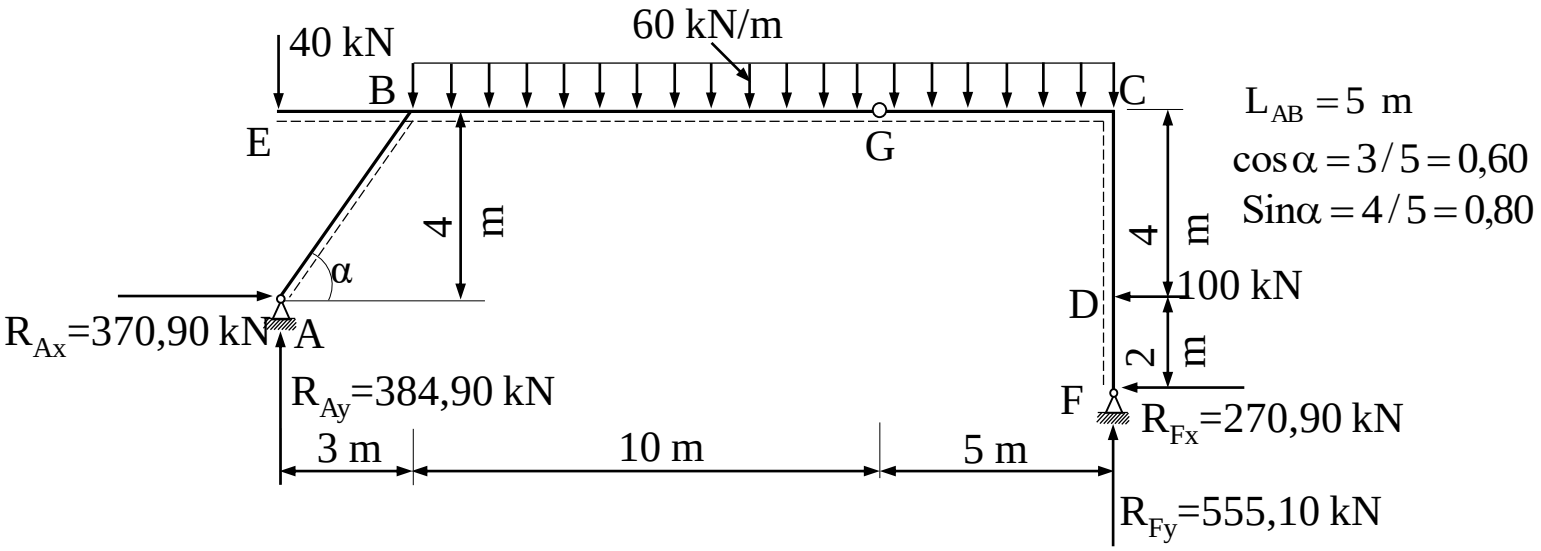
Estimate support reactions and draw internal force diagrams of the given system.

Solution

There are 4 support reaction on this system, thus these reactions are obtained by 3 equilibrium equations and 1 hinge condition. System is statically determinate.

Memeber lengths and their trigonometric identities:

$L_{AB} = \sqrt{3^2 + 4^2} = 5 \text{ m}$ $\quad \cos\alpha = 3/5 = 0{,}60$ $\quad \text{Sin}\alpha = 4/5 = 0{,}80$



Support reactions

For ABG

$$\sum M_G = 0 \qquad R_{Ay} \cdot 13{,}0 - R_{Ax} \cdot 4{,}0 - 40 \times 13{,}0 - 60 \times 10 \times 5{,}0 = 0$$

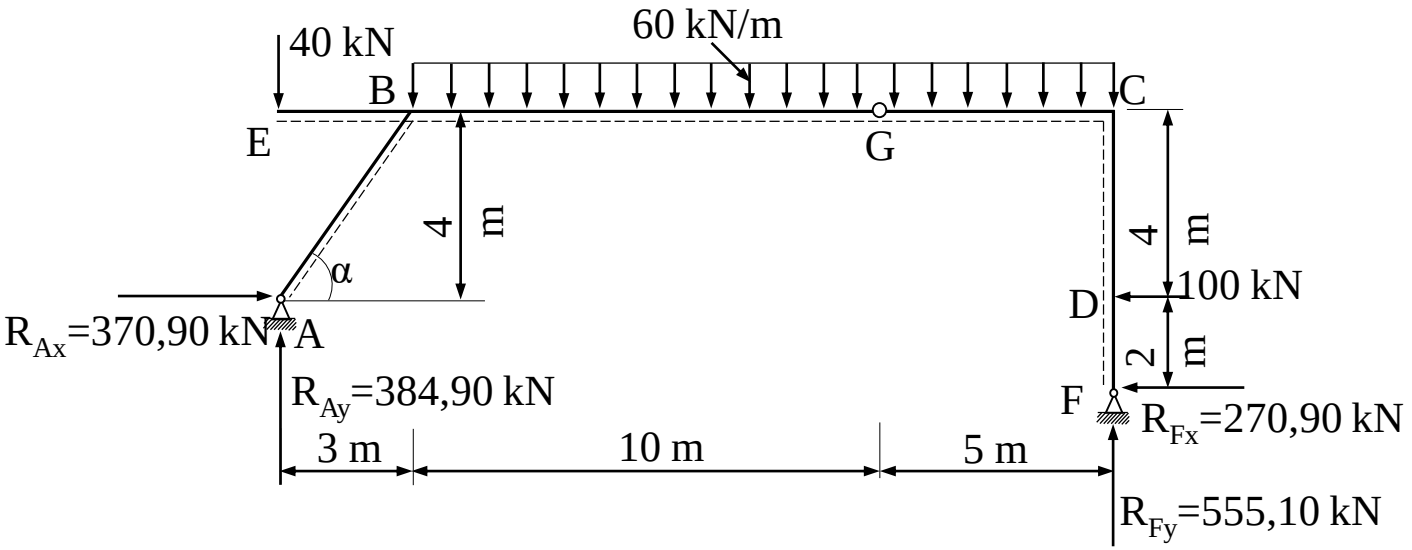
$$13R_{Ay} - 4R_{Ax} - 3520 = 0$$

For whole system $\sum M_F = 0$

$$R_{Ay} \cdot 18{,}0 + R_{Ax} \cdot 2{,}0 - 40 \times 18{,}0 - 60 \times 15 \times 7{,}50 - 100 \times 2{,}0 = 0$$

$$18R_{Ay} + 2R_{Ax} - 7670 = 0$$

$$R_{Ax} = 370{,}90 \ \text{kN} \qquad R_{Ay} = 384{,}90 \ \text{kN}$$



For GCF $\sum M_G = 0$ Hinge condition

$$-R_{Fy} \cdot 5{,}0 + R_{Fx} \cdot 6{,}0 + 100 \times 4{,}0 + 60 \times 5{,}0 \times 2{,}5 = 0$$

$$-5R_{Fy} + 6R_{Fx} + 1150 = 0$$

Whole system $\sum M_A = 0$

$$-R_{Fy} \cdot 18{,}0 + R_{Fx} \cdot 2{,}0 + 40 \times 0{,}0 + 100 \times 0{,}0 + 60 \times 15{,}0 \times 10{,}5 = 0$$

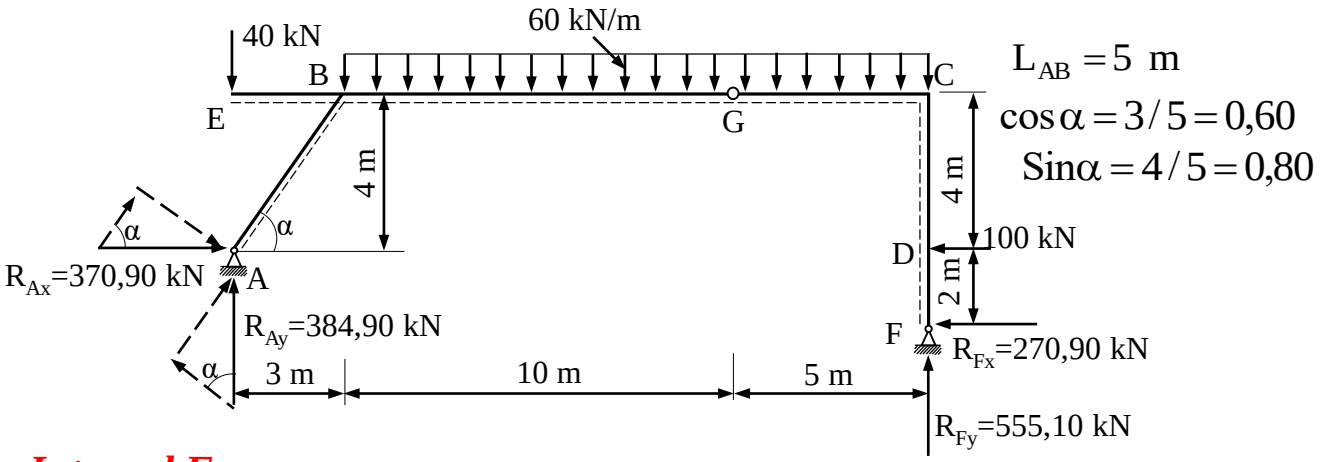
$$-18R_{Fy} + 2R_{Fx} + 9450 = 0$$

$$R_{Fx} = 270{,}90 \ \text{kN} \qquad R_{Fy} = 555{,}10 \ \text{kN}$$

Check the results:

$$\sum P_x = 370{,}90 - 100 - 270{,}90 = 0$$

$$\sum P_y = 384{,}90 - 40 - 900 + 555{,}10 = 0$$



Internal Forces

$$N_{AB} = N_{BA} = -370{,}90 \cdot \cos\alpha - 384{,}90 \cdot \sin\alpha$$

$$N_{AB} = N_{BA} = -370{,}90 \times 0{,}6 - 384{,}90 \times 0{,}80 = -530{,}46 \ \text{kN}$$

$$V_{AB} = V_{BA} = -370{,}90 \cdot \sin\alpha + 384{,}90 \cdot \cos\alpha$$

$$V_{AB} = V_{BA} = -370{,}90 \times 0{,}8 + 384{,}90 \times 0{,}60 = -65{,}78 \ \text{kN}$$

$$M_{AB} = 0 \qquad M_{BA} = 384{,}90 \times 3{,}0 - 370{,}90 \times 4{,}0 = -328{,}90 \ \text{kNm}$$

$$N_{EB} = N_{BE} = 0 \qquad V_{EB} = V_{BE} = -40 \ \text{kN} \qquad M_{EB} = 0$$

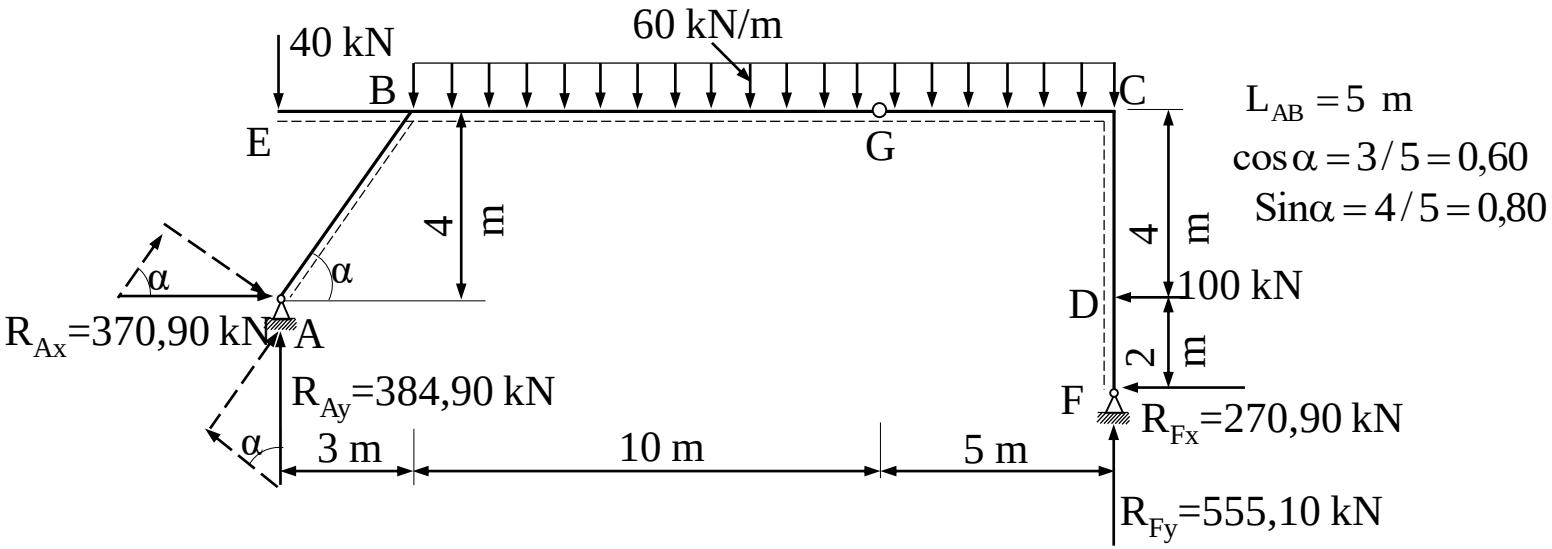
$$M_{BE} = -40 \times 3{,}0 = -120 \ \text{kNm}$$

$$N_{BG} = N_{GB} = N_{GC} = N_{CG} = -370{,}90 \ kN$$

$$V_{BG} = 384{,}90 - 40 = 344{,}90 \ \text{kN}$$

$$M_{BG} = 384{,}90 \times 3{,}0 - 370{,}90 \times 4{,}0 - 40 \times 3 = -448{,}90 \ \text{kNm}$$

$$V_{GB} = V_{GC} = -555{,}10 + 60 \times 5{,}0 = -255{,}10 \ \text{kN}$$



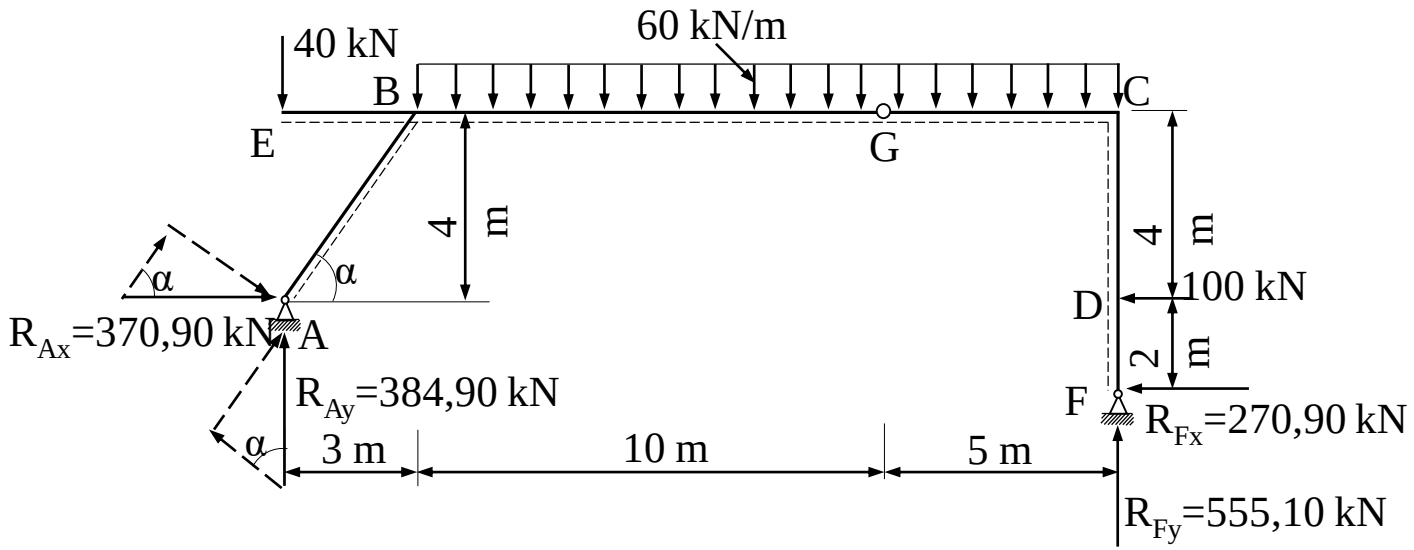
$V_{BG} = 344{,}90$ kN $\qquad M_{BG} = -448{,}90$ kNm

$V_{GB} = -255{,}10$ kN

Since the sign of the shear forces at B and G ends of BG frame changes, shear force is zero at any location and moment is maximum at that point.

$$x_0 = V_{BG} / q = 344{,}90 / 60 = 5{,}75 \text{ m}$$

$$M_{enb.} = M_{BG} + V_{BG}^2 /(2q) = -448{,}90 + 344{,}9^2 /(2 \times 60) = 542{,}40 \text{ kNm}$$



$$V_{CG} = -555{,}1 \ \text{kN}$$

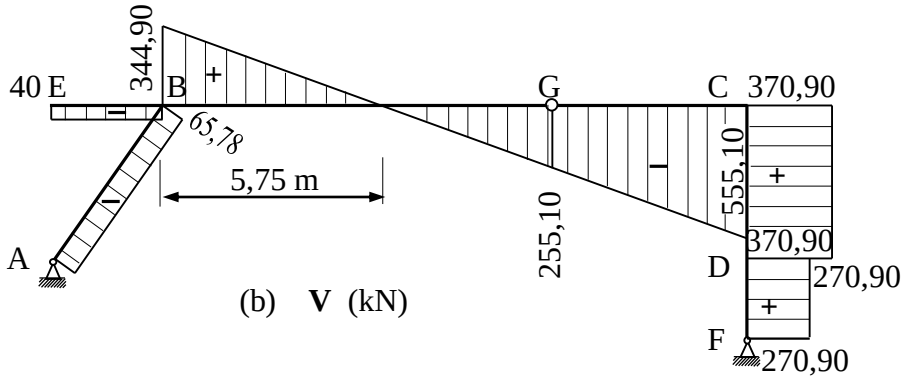
$$M_{CG} = M_{CD} = -270{,}90 \times 6{,}0 - 100 \times 4{,}0 = -2025{,}40 \ \text{kNm}$$

$$N_{CD} = N_{DC} = N_{DF} = N_{FD} = -555{,}10 \ \text{kN}$$

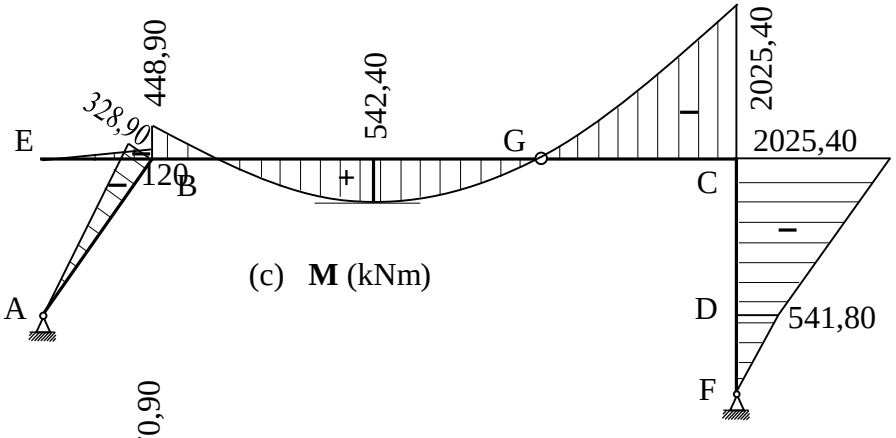
$$V_{CD} = V_{DC} = 270{,}90 + 100 = 370{,}90 \ \text{kN}$$

$$M_{DC} = M_{DF} = -270{,}90 \times 2{,}0 = -541{,}80 \ \text{kNm}$$

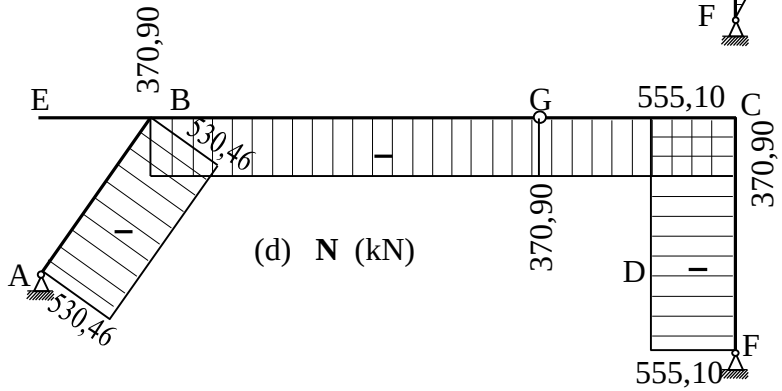
$$V_{DF} = V_{FD} = 270{,}90 \ \text{kN} \qquad M_{FD} = 0$$



(b) **V** (kN)

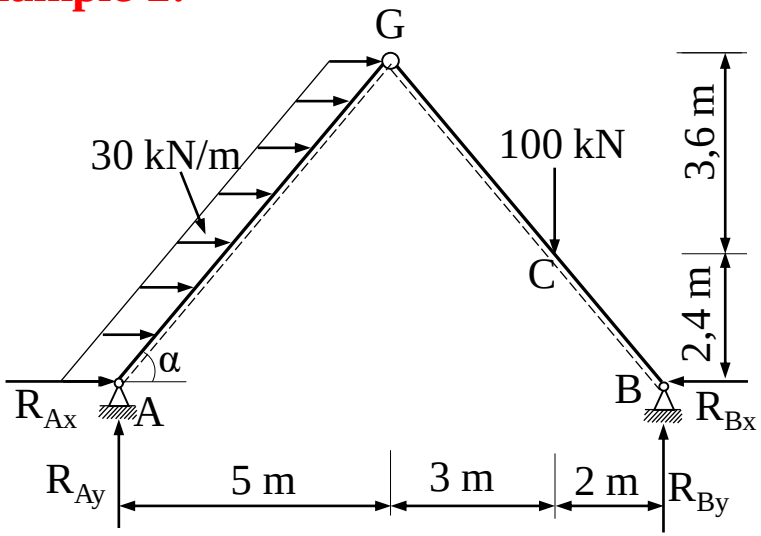


(c) **M** (kNm)



(d) **N** (kN)

Example 2:



Estimate support reactions and draw internal force diagrams of the given system.

Solution

$$L_{AG} = L_{BG} = \sqrt{5^2 + 6^2} = 7{,}81 \text{ m}$$

$$\sin\alpha = 6/7{,}81 = 0{,}7682$$

$$\cos\alpha = 5/7{,}81 = 0{,}6402$$

Support Reactions

Resultant force on AG member

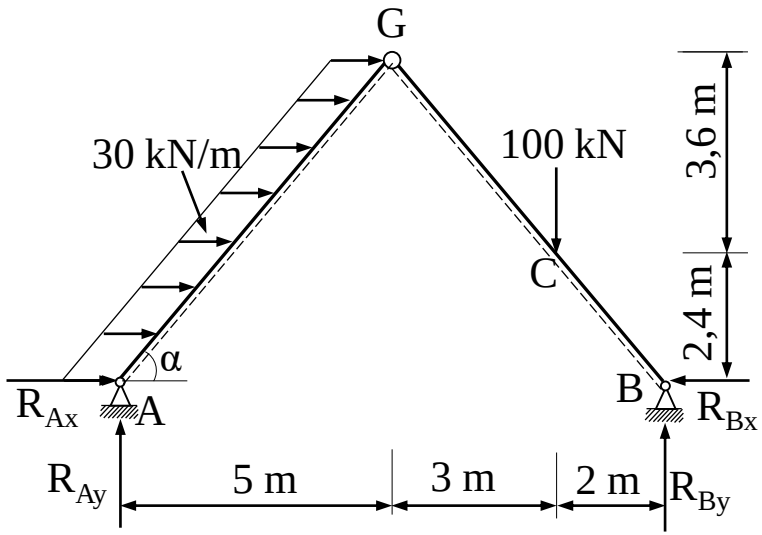
$$Q_{AG} = 30 \times 7{,}81 = 234{,}30 \text{ kN}$$

For whole system

$$\sum M_B = 0 \longrightarrow R_{Ay} = (-234{,}30 \times 3{,}0 + 100 \times 2{,}0)/10 = -50{,}29 \text{ kN}$$

$$\sum M_A = 0 \longrightarrow R_{By} = (234{,}30 \times 3{,}0 + 100 \times 8{,}0)/10 = 150{,}29 \text{ kN}$$

Check: $\sum P_y = -50{,}29 - 100 + 150{,}29 = 0$



$L_{AG} = 7{,}81 \text{ m}$

$\sin\alpha = 6/7{,}81 = 0{,}7682$

$\cos\alpha = 5/7{,}81 = 0{,}6402$

$Q_{AG} = 234{,}30 \text{ kN}$

Horizontal support reactions are obtained by writing the hinge condition on AG and BG systems, separately.

For AG system $\sum M_G = 0$

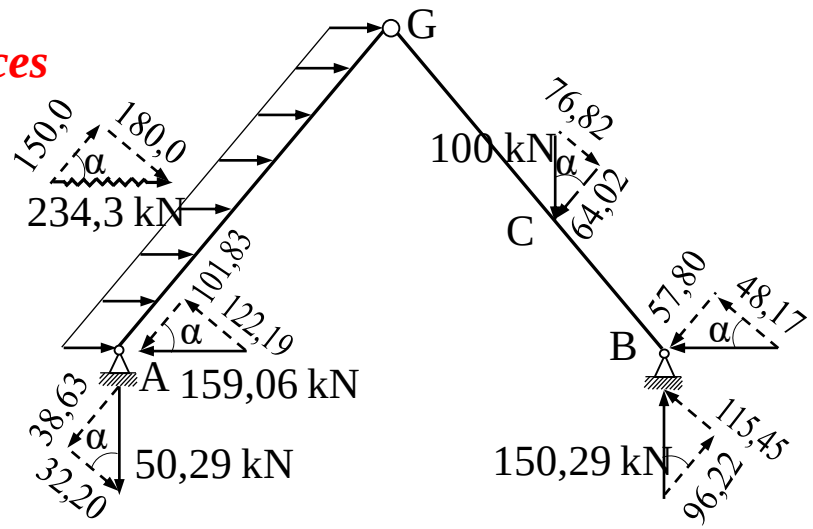
$$R_{Ax} = (-50{,}29 \times 5{,}0 - 234{,}30 \times 3{,}0)/6 = -159{,}06 \text{ kN}$$

For GB system $\sum M_G = 0$

$$R_{Bx} = (150{,}29 \times 5{,}0 - 100 \times 3{,}0)/6 = 75{,}24 \text{ kN}$$

Check $\sum P_x = -159{,}06 + 234{,}30 - 75{,}24 = 0$

Internal Forces



$L_{AG} = 7{,}81 \text{ m}$

$\sin\alpha = 6/7{,}81 = 0{,}7682$

$\cos\alpha = 5/7{,}81 = 0{,}6402$

Resultant forces of the external loads, which are parallel and perpendicular to inclined members, are estimated to obtain the axial and shear forces of the members

$$N_{AG} = 38{,}63 + 101{,}83 = 140{,}46 \text{ kN}$$

$$V_{AG} = -32{,}20 + 122{,}19 = 89{,}99 \text{ kN} \qquad M_{AG} = 0$$

$$N_{GA} = 38{,}63 + 101{,}83 - 150 = -9{,}54 \text{ kN}$$

$$V_{GA} = -32{,}20 + 122{,}19 - 180 = -90{,}01 \text{ kN} \qquad M_{GA} = 0$$

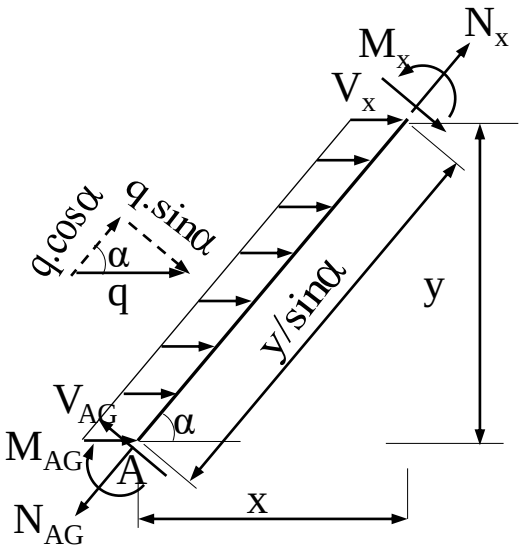
$$N_{BC} = N_{CB} = -115{,}45 - 48{,}17 = -163{,}62 \text{ kN}$$

$$V_{BC} = V_{CB} = -96{,}22 + 57{,}80 = -38{,}42 \text{ kN} \qquad M_{BC} = 0$$

$$M_{CB} = M_{CG} = 150{,}29 \times 2{,}0 - 75{,}24 \times 2{,}4 = 120{,}00 \text{ kNm} \qquad M_{GC} = 0$$

$$N_{CG} = N_{GC} = -115{,}45 - 48{,}17 + 76{,}82 = -86{,}80 \text{ kN}$$

$$V_{CG} = V_{GC} = -96{,}22 + 57{,}80 + 64{,}02 = 25{,}60 \text{ kN}$$



Since the sign of the shear forces at A and G ends of AG frame changes , shear force is zero at any location and moment is maximum at that point.

$$V_x = V_{AG} - q \cdot \sin\alpha \cdot (y / \sin\alpha) = 0$$
$$V_x = V_{AG} - q \cdot y = 0$$

$y_0 = V_{AG} / q$ Height of the zero shear point from the support A

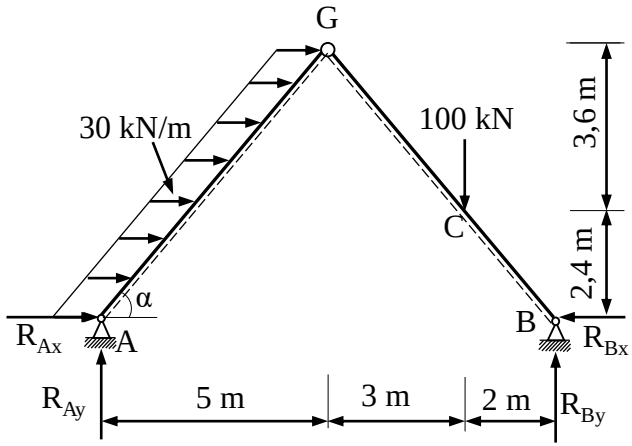
$$x_0 = y_0 / tg\alpha$$

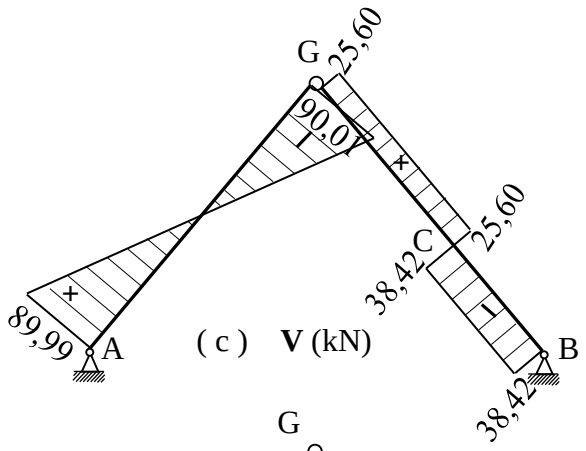
$$M_x = M_{AG} + V_{AG} \cdot (y / \sin\alpha) - q \cdot \sin\alpha \cdot (y / \sin\alpha)(y / 2\sin\alpha)$$
$$M_x = M_{AG} + V_{AG} \cdot y / \sin\alpha - q \cdot y^2 / 2\sin\alpha$$

$y = y_0 = V_{AG} / q$ $M_{enb.} = M_{AG} + V_{AG}^2 / (2q \cdot \sin\alpha)$

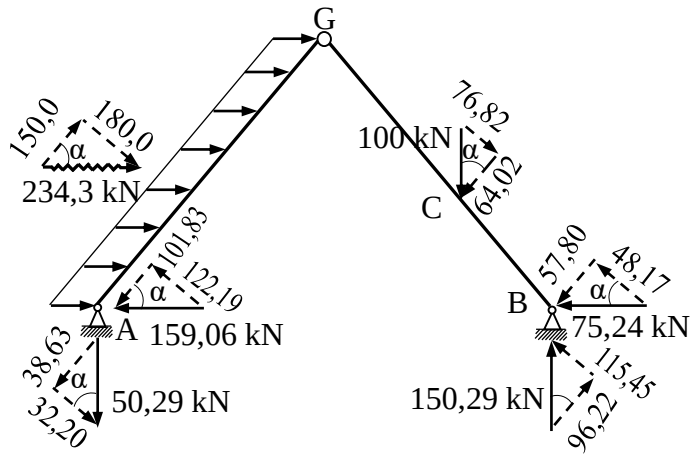
$y_0 = 89{,}99 / 30 = 3{,}0$ m $x_0 = 3{,}0 / 1{,}2 = 2{,}50$ m

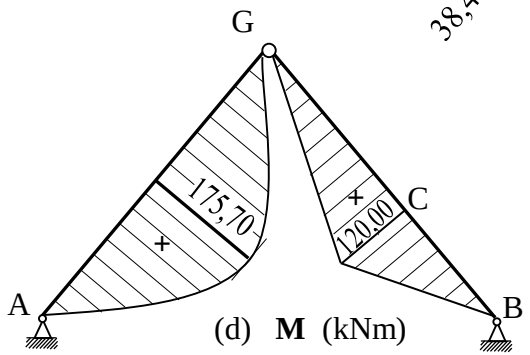
$M_{enb.} = 0 + 89{,}99^2 / (2 \times 30 \times 0{,}7682) = 175{,}70$ kNm



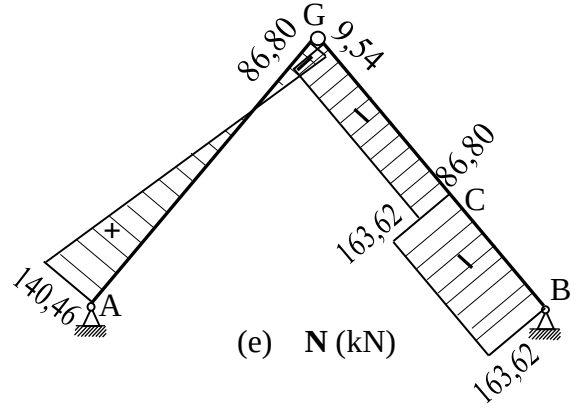


(c) **V** (kN)





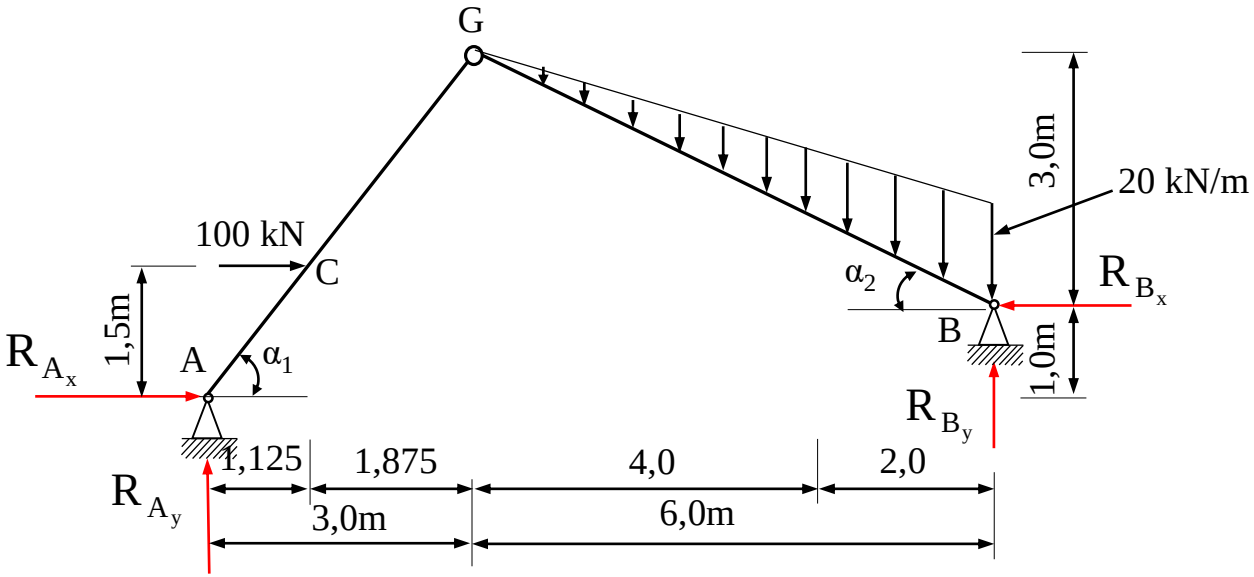
(d) **M** (kNm)



(e) **N** (kN)

Example 3:

Estimate support reactions and draw internal force diagrams of the given system.

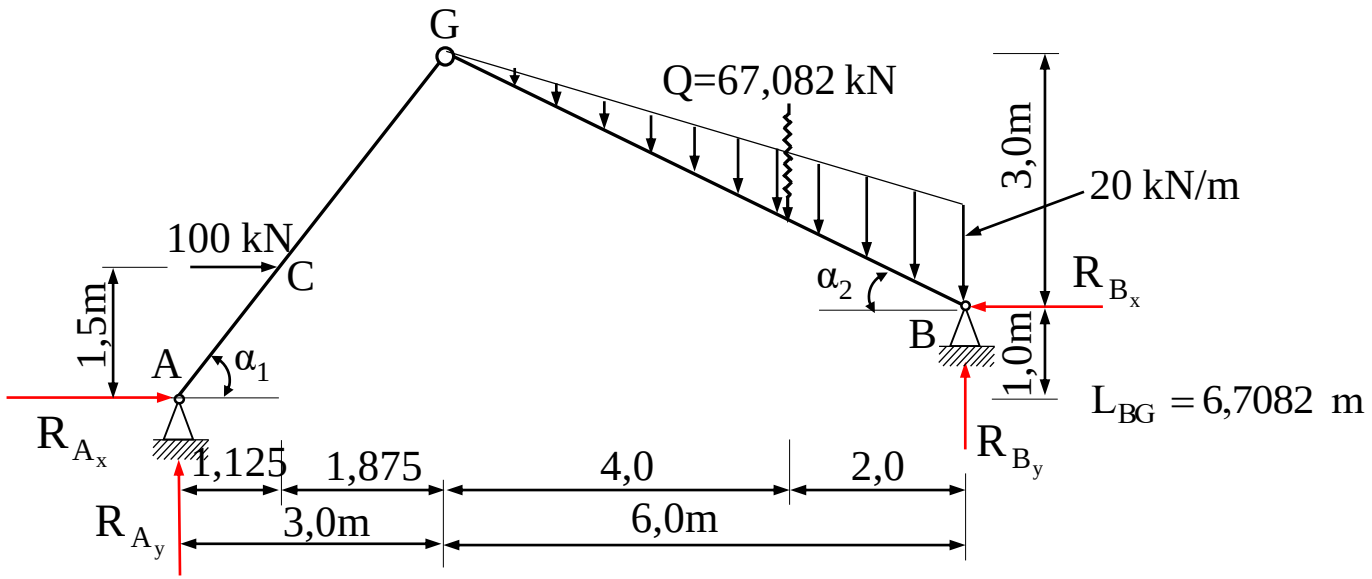


Solution:

Horizontal distance between A and C is 1,125 m.

Length of BG member:

$$L_{BG} = \sqrt{3^2 + 6^2} = 6{,}7082 \text{ m}$$



Resultant force on BG member

$$Q = 0,5 \times 20 \times 6,7082 = 67,082 \text{ kN}$$

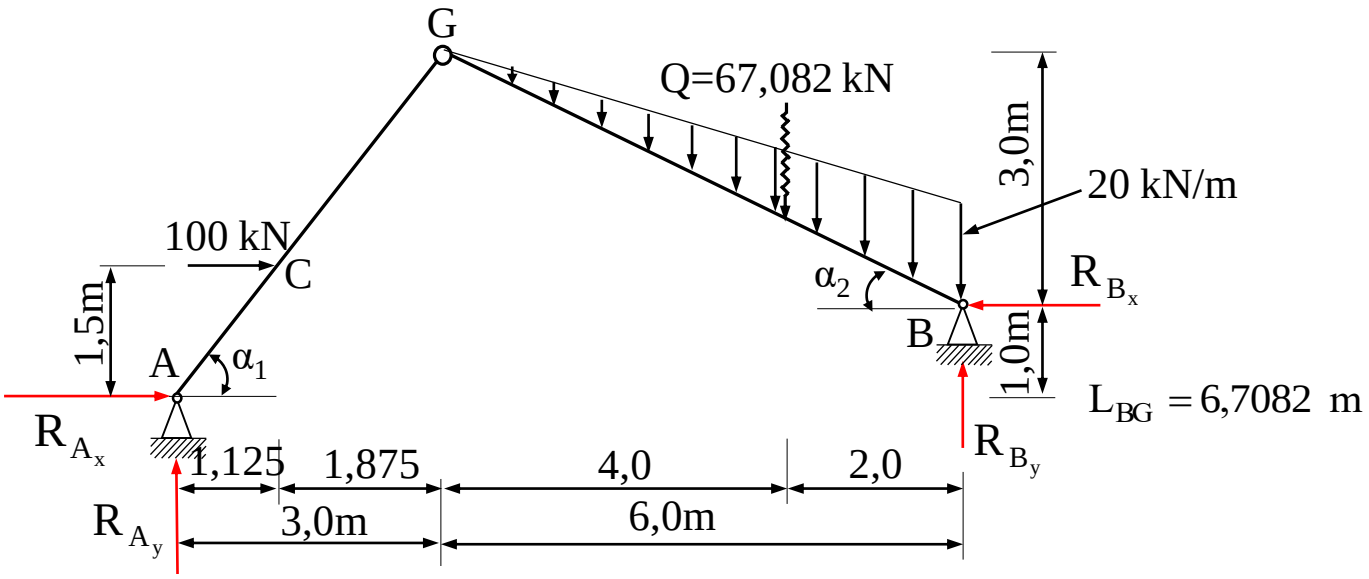
For AG system: $\sum M_G = 0$

$$R_{A_y}.3,0 - R_{A_x} \cdot 4,0 - 100 \times 2,50 = 0 \longrightarrow 3R_{A_y} - 4R_{A_x} = 250$$

For whole system: $\sum M_B = 0$

$$R_{A_y}.9,0 - R_{A_x} \cdot 1,0 + 100 \times 0,5 - 67,082 \times 2,0 = 0 \longrightarrow 9R_{A_y} - R_{A_x} = 84,164$$

$$R_{A_x} = -60,531 \text{ kN} \quad R_{A_y} = 2,626 \text{ kN}$$



For GB system: $\sum M_G = 0$

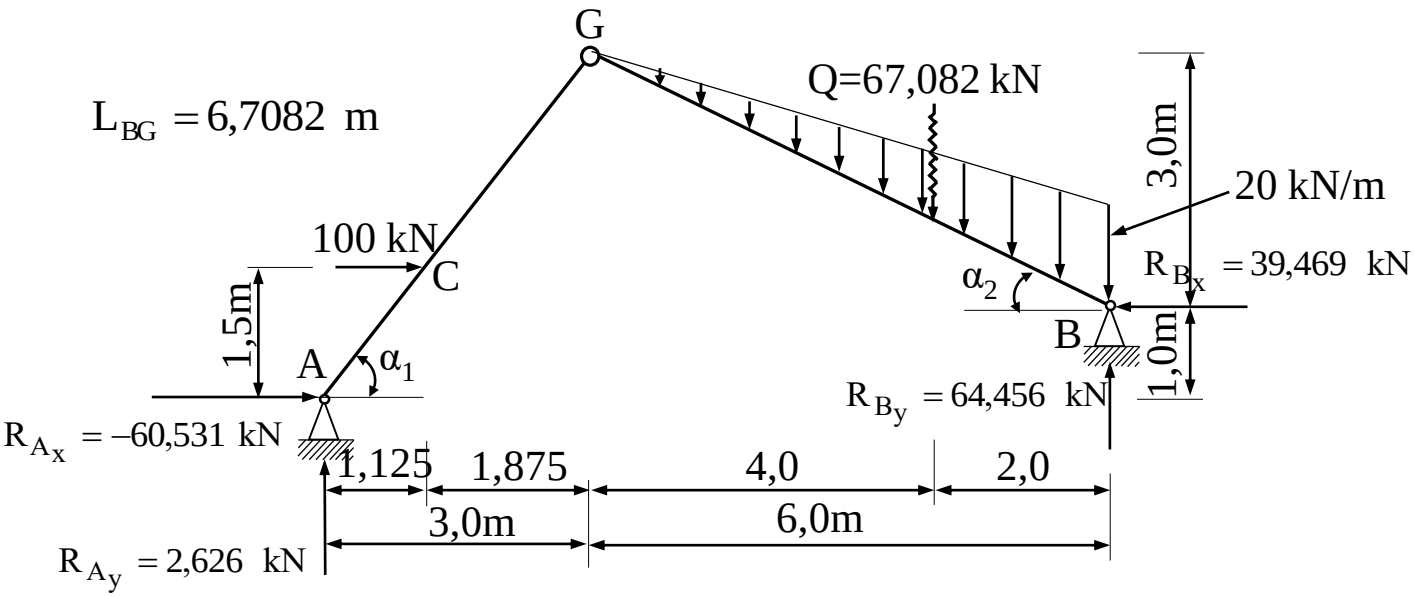
$$-R_{B_y}.6{,}0 + R_{B_x} \cdot 3{,}0 + 67{,}082 \times 4{,}0 = 0 \longrightarrow -2R_{B_y} + R_{B_x} = -89{,}443$$

For whole system: $\sum M_A = 0$

$$-R_{B_y}.9{,}0 - R_{B_x} \cdot 1{,}0 + 100 \times 1{,}5 + 67{,}082 \times 7{,}0 = 0$$

$$\longrightarrow -9R_{B_y} - R_{B_x} = -619{,}574$$

$$R_{B_x} = 39{,}469 \text{ kN} \qquad R_{B_y} = 64{,}456 \text{ kN}$$



Check the results:

Fro whole system:

$$\sum P_x = R_{A_x} - R_{B_x} + 100 = -60{,}531 - 39{,}469 + 100 = 0$$

$$\sum P_y = R_{A_y} + R_{B_x} - 67{,}082 = 2{,}626 + 64{,}456 - 67{,}082 = 0$$

The estimated support reactions are correct

Member lenghts and their trigonometric identities:

$$L_{BG} = \sqrt{3^2 + 6^2} = 6{,}7082 \text{ m}$$

$$\cos\alpha_1 = 3/5 = 0{,}60$$

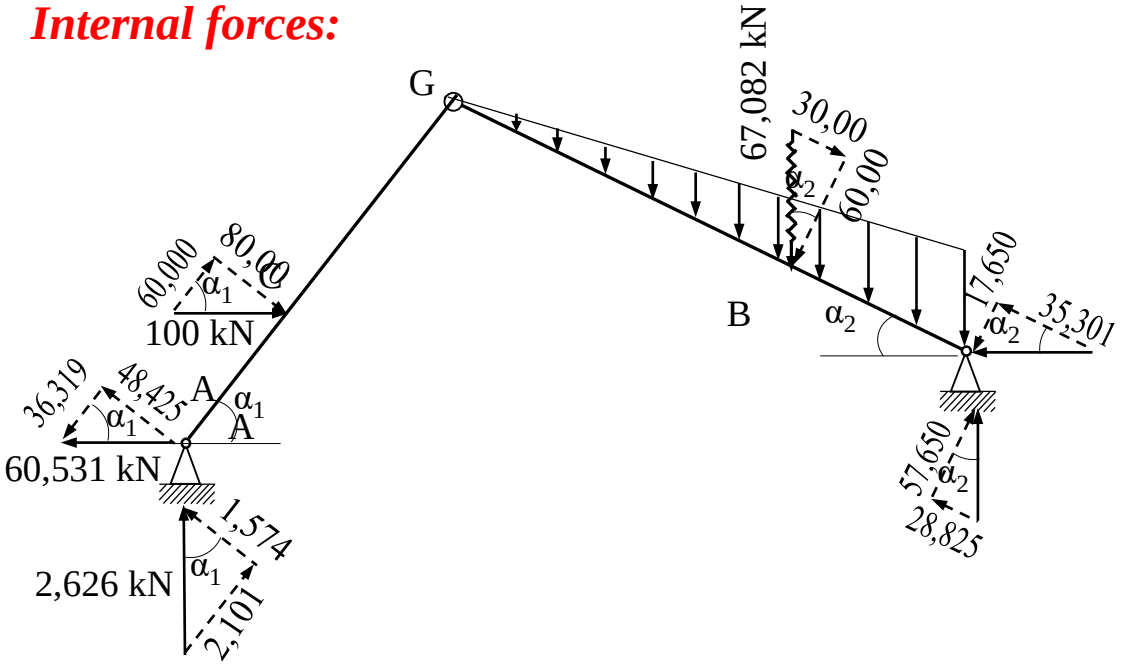
$$\cos\alpha_2 = 6/6{,}7082 = 0{,}8944$$

$$L_{AG} = \sqrt{4^2 + 3^2} = 5{,}0 \text{ m}$$

$$\sin\alpha_1 = 4/5 = 0{,}80$$

$$\sin\alpha_2 = 3/6{,}7082 = 0{,}4472$$

Internal forces:



$L_{BG} = 6{,}7082 \text{ m}$

$L_{AG} = 5{,}0 \text{ m}$

$\cos\alpha_1 = 0{,}60$

$\sin\alpha_1 = 0{,}80$

$\cos\alpha_2 = 0{,}8944$

$\sin\alpha_2 = 0{,}4472$

AC and CG members

$N_{AC} = N_{CA} = -2{,}101 + 36{,}319 = 34{,}218 \text{ kN}$

$V_{AC} = V_{CA} = 48{,}425 + 1{,}575 = 50{,}00 \text{ kN}$ $\qquad M_{AC} = 0$

$M_{CA} = M_{CG} = 60{,}531 \times 1{,}5 + 2{,}626 \times 1{,}125 = 92{,}75 \text{ kNm}$

$N_{CG} = N_{GC} = -2{,}101 + 36{,}319 - 60{,}00 = -25{,}782 \text{ kN}$

$V_{CG} = V_{GC} = 48{,}425 + 1{,}575 - 80{,}00 = -30{,}00 \text{ kN}$ $\qquad M_{GC} = 0$

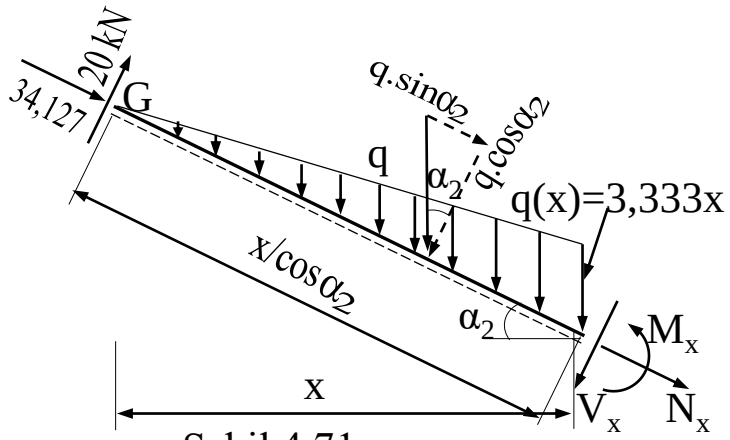
BG member

$N_{BG} = -28{,}825 - 35{,}301 = -64{,}126 \text{ kN}$

$V_{BG} = -57{,}650 + 17{,}650 = -40{,}00 \text{ kN}$ $\qquad M_{BG} = 0$

$N_{GB} = -28{,}825 - 35{,}301 + 30 = -34{,}126 \text{ kN}$

$V_{GB} = -57{,}650 + 17{,}650 + 60 = 20{,}00 \text{ kN}$ $\qquad M_{GB} = 0$



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$V_{BG} = -40{,}00$ kN

$M_{BG} = 0$

$V_{GB} = 20{,}00$ kN

$M_{GB} = 0$

Since the sign of the shear forces at B and G ends of BG frame changes, shear force is zero at any location and moment is maximum at that point.

The function of triangular distributed load: $q(x) = (x/6)q = (20/6)x = 3{,}333x$

$$V_x = 20 - 0{,}5 \cdot q(x).\cos\alpha_2.(x/\cos\alpha_2)$$

$$V_x = 20 - 1{,}667x^2 = 0 \qquad x_0 = \sqrt{20/1{,}667} = 3{,}464 \text{ m}$$

$$M_x = 20 \cdot x/\cos\alpha_2 - 0{,}5 \cdot q(x) \cdot \cos\alpha_2 \cdot (x/\cos\alpha_2) \cdot x/(3 \cdot \cos\alpha_2)$$

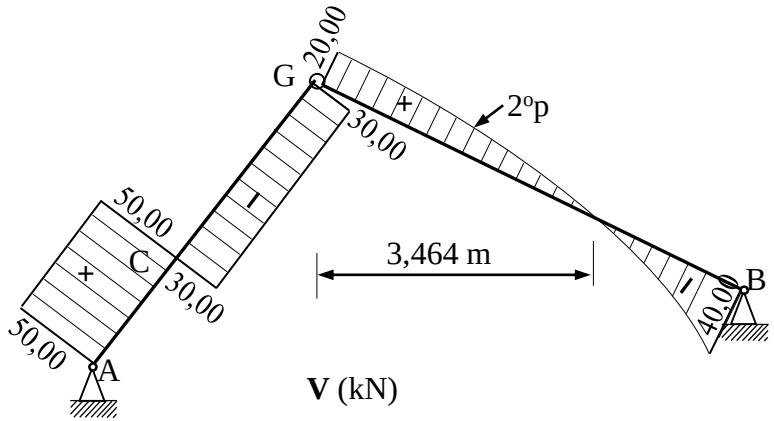
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$$M_x = 20 \cdot x/\cos\alpha_2 - 0{,}5 \cdot q(x) \cdot (x/\cos\alpha_2) \cdot x/3$$

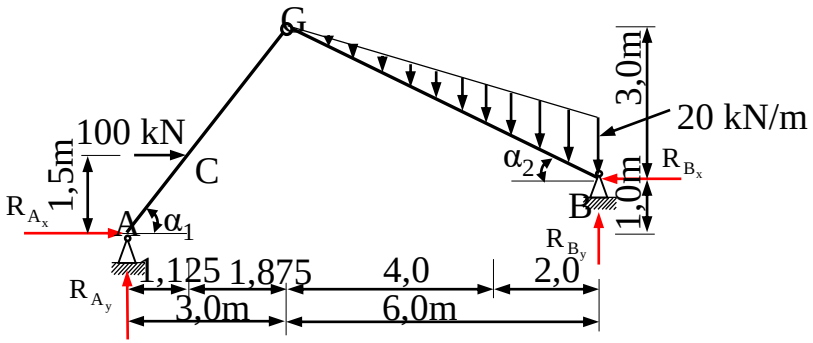
$$M_x = (20 \cdot x - 0{,}555x^3)/\cos\alpha_2$$

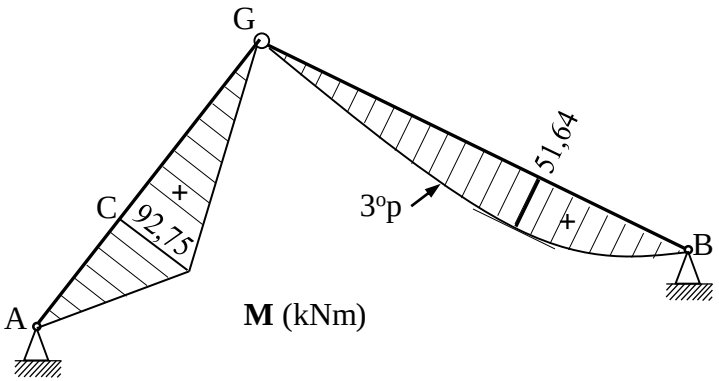
Substitute 3,464 m instead of x

$$M_{enb.} = (20 \times 3{,}464 - 0{,}555 \times 3{,}464^3)/0{,}8944 = 51{,}64 \text{ kNm}$$

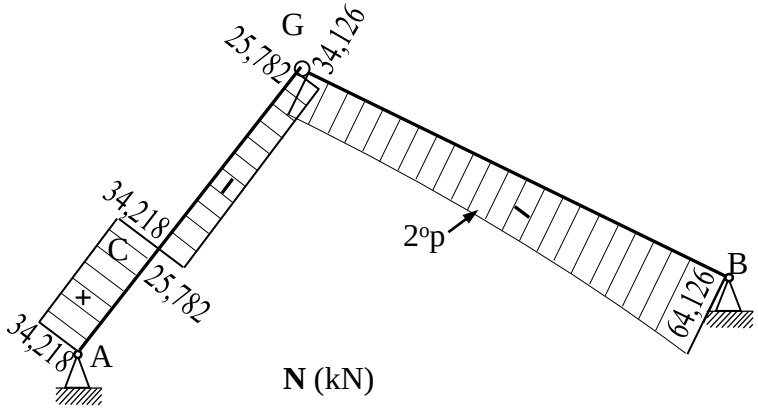

G
20,00
2ºp
30,00
50,00
C
30,00
3,464 m
50,00
A
40,00
B

V (kN)


G
3,0m
20 kN/m
100 kN
1,5m
C
R_{B_x}
α_2
R_{A_x}
A
α_1
B
1,0m
R_{B_y}
R_{A_y}
1,125
1,875
4,0
2,0
3,0m
6,0m

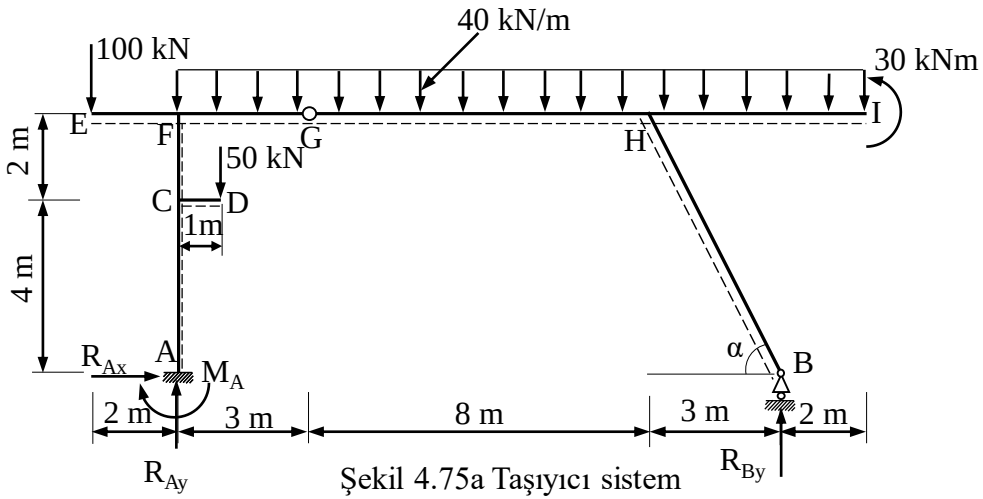

G
51,64
C
92,75
3ºp
B
A

M (kNm)


G
25,782
34,126
34,218
C
25,782
2ºp
B
64,126
34,218
A

N (kN)

Example 4:



Şekil 4.75a Taşıyıcı sistem

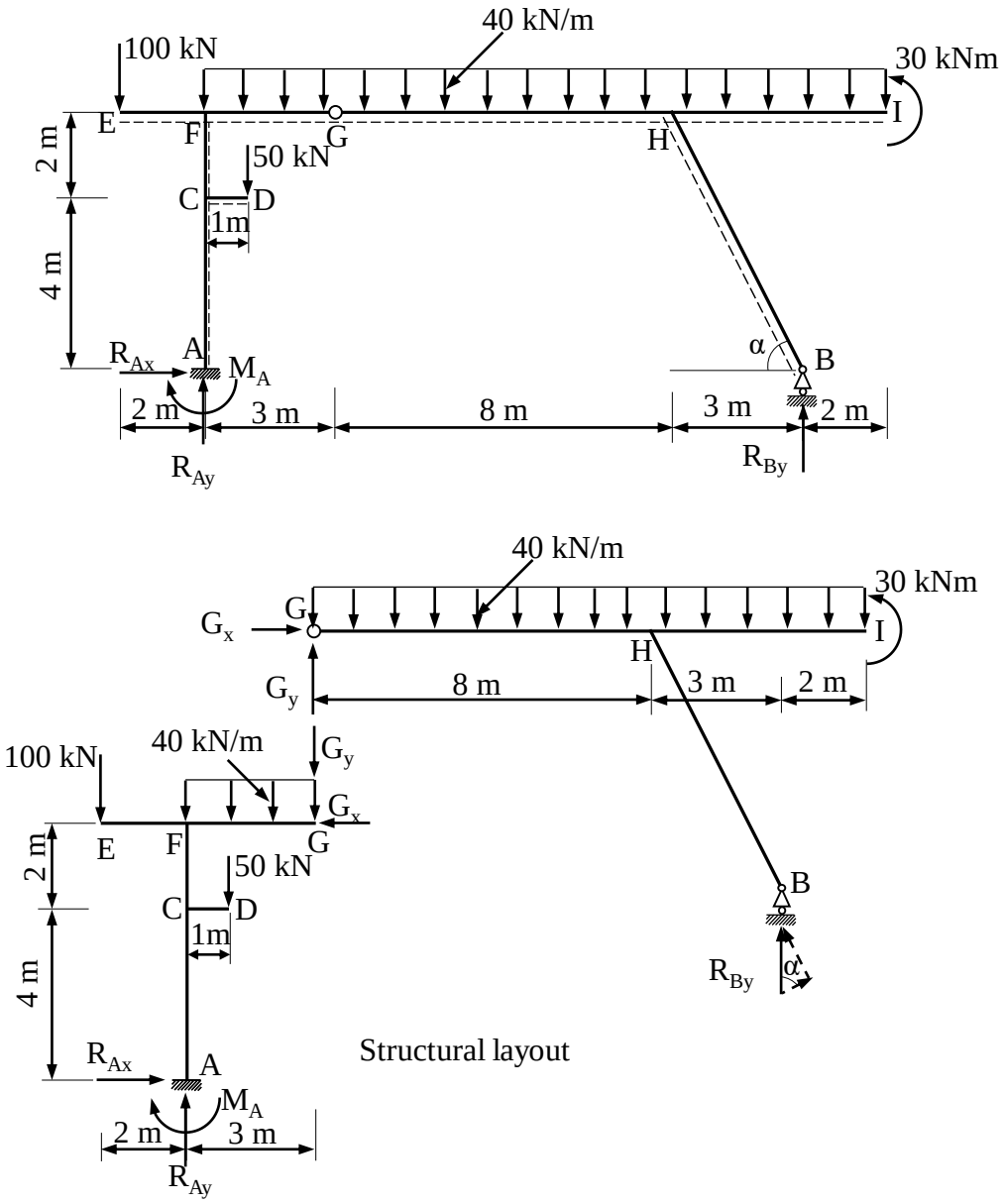
Estimate support reactions and draw internal force diagrams of the given system.

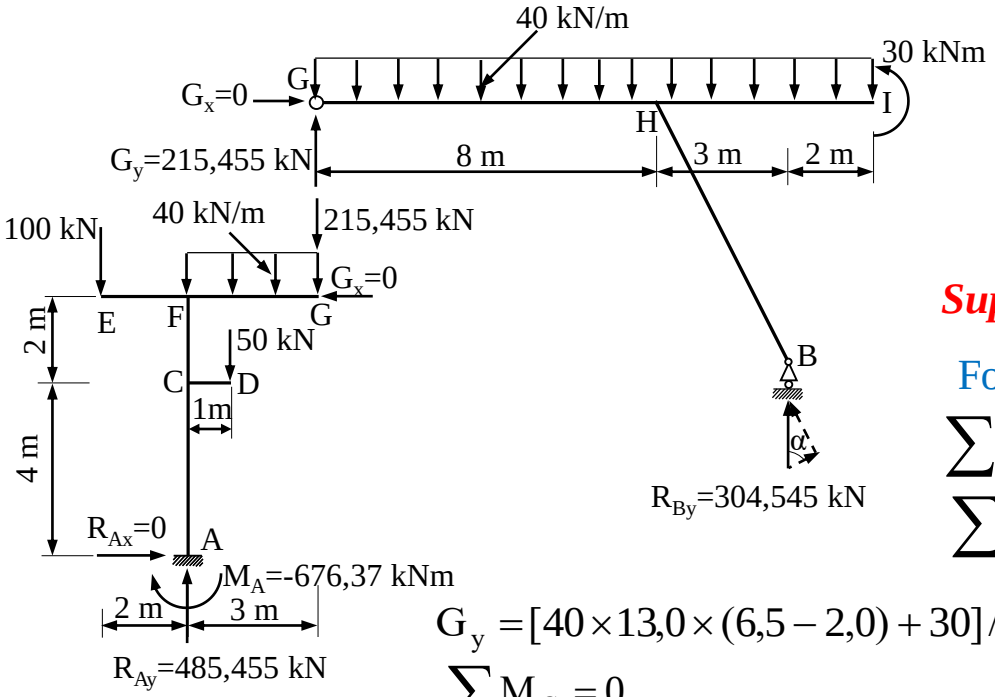
Solution:

Member lenghts and their trigonometric identities:

$$L_{BH} = \sqrt{6^2 + 3^2} = 6{,}71 \text{ m}$$

$$\cos\alpha = 3/6{,}71 = 0{,}447 \qquad \sin\alpha = 6/6{,}71 = 0{,}894$$


40 kN/m
100 kN
30 kNm
E
F
G
H
I
2 m
50 kN
C
D
1m
4 m
R_{Ax}
A
M_A
α
B
2 m
3 m
8 m
3 m
2 m
R_{Ay}
R_{By}
G_x
G_y
Structural layout



Support Reactions:

For GHB system

$$\sum P_x = 0 \quad G_x = 0$$

$$\sum M_B = 0$$

$$G_y = [40 \times 13{,}0 \times (6{,}5 - 2{,}0) + 30]/11 = 215{,}455 \text{ kN}$$

$$\sum M_G = 0$$

$$R_{By} = (40 \times 13{,}0 \times 6{,}5 - 30]/11 = 304{,}545 \text{ kN}$$

Check the results: $\sum P_y = 215{,}455 - 520{,}000 + 304{,}545 = 0$

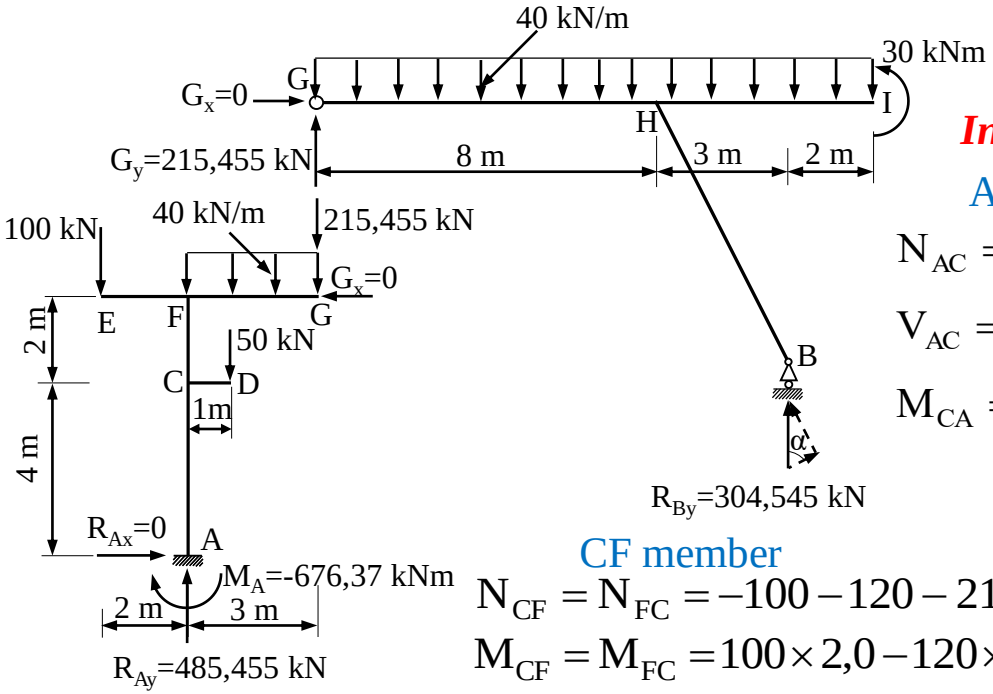
For AFG system

$$\sum P_x = 0 \qquad R_{Ax} = 0$$

$$\sum P_y = 0 \qquad R_{Ay} = 215{,}455 + 40 \times 3{,}0 + 100 + 50 = 485{,}455 \text{ kN}$$

$$\sum M_A = 0$$

$$M_A = -215{,}455 \times 3{,}0 - 40 \times 3{,}0 \times 1{,}5 + 100 \times 2{,}0 - 50 \times 1{,}0 = -676{,}37 \text{ kNm}$$



Internal Forces

AC member

$$N_{AC} = N_{CA} = -485{,}455 \text{ kN}$$

$$V_{AC} = V_{CA} = 0$$

$$M_{CA} = M_{CA} = -676{,}37 \text{ kNm}$$

CF member

$$N_{CF} = N_{FC} = -100 - 120 - 215{,}455 = -435{,}455 \text{ kN},$$

$$M_{CF} = M_{FC} = 100 \times 2{,}0 - 120 \times 1{,}5 - 215{,}455 \times 3{,}0$$

$$M_{CF} = M_{FC} = -626{,}37 \text{ kNm} \qquad V_{CF} = V_{FC} = 0$$

CD member

$$N_{CD} = N_{DC} = 0 \qquad V_{CD} = V_{DC} = 50 \text{ kN}$$

$$M_{CD} = -50 \times 1{,}0 = -50{,}00 \text{ kNm} \qquad M_{DC} = 0$$

EF member

$$N_{EF} = N_{FE} = 0 \qquad V_{EF} = V_{FE} = -100 \text{ kN}$$

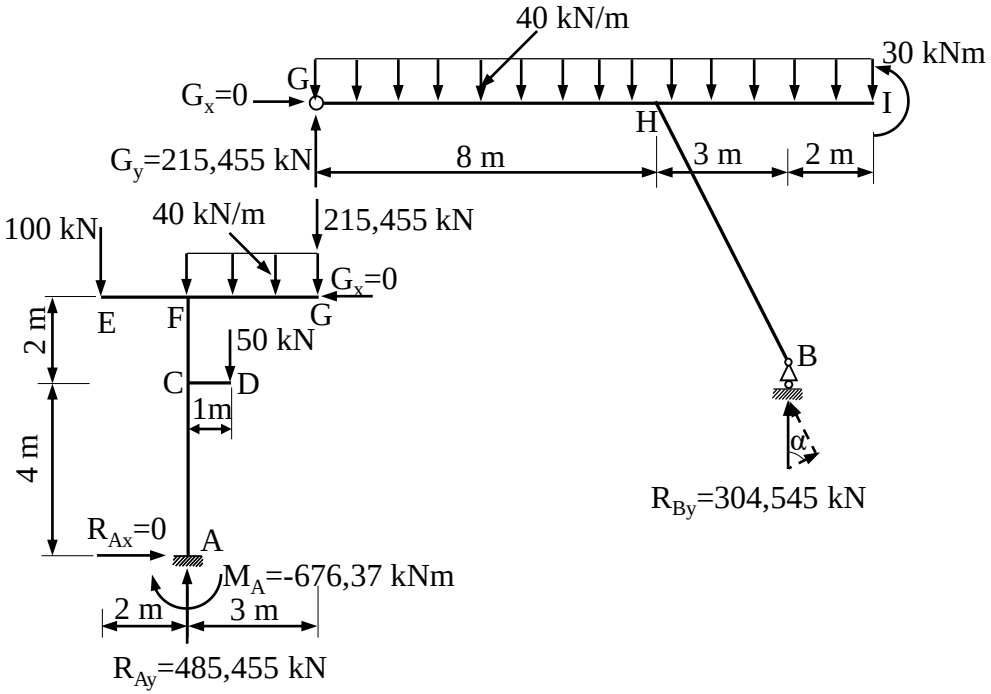
$$M_{FE} = -100 \times 2{,}0 = -200{,}00 \text{ kNm} \qquad M_{EF} = 0$$

FG member

$$N_{FG} = N_{GF} = 0 \qquad V_{FG} = 215{,}455 + 120 = 335{,}455 \text{ kN}$$

$$V_{GF} = 215{,}455 \text{ kN}$$

$$M_{FG} = -215{,}455 \times 3{,}0 - 120 \times 1{,}5 = -826{,}37 \text{ kNm} \qquad M_{GF} = 0$$



GH member

$$N_{GH} = N_{HG} = 0$$

$$V_{GH} = V_{GF} = 215{,}455 \ \ kN$$

$$M_{GH} = M_{GF} = 0$$

$$V_{HG} = 215{,}455 - 320 = -104{,}445 \ \ kN$$

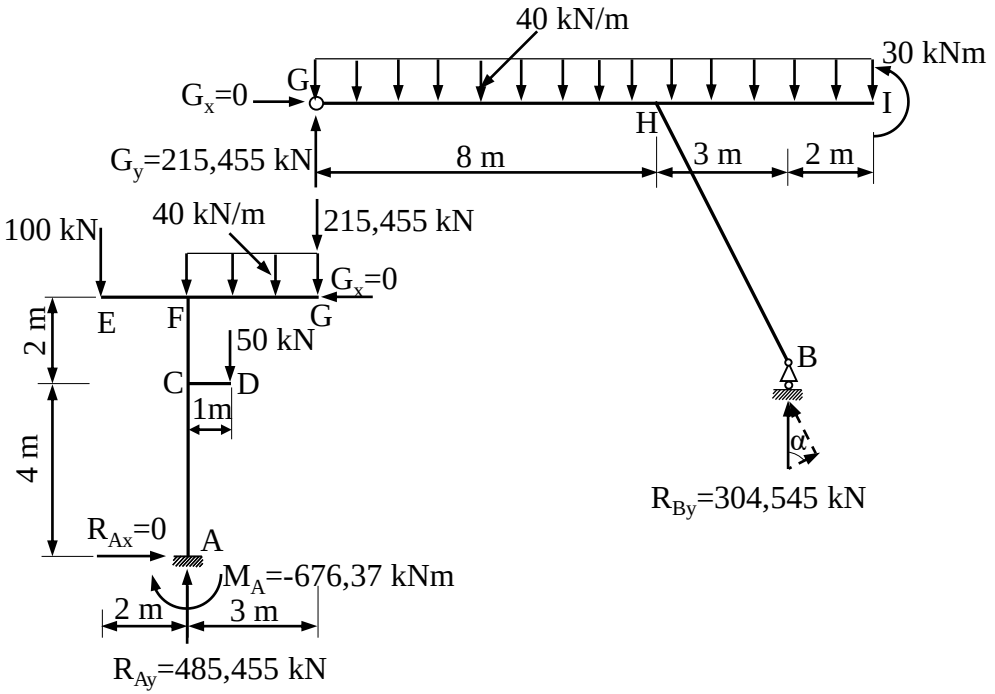
$$M_{HG} = 215{,}455 \times 8{,}0 - 320 \times 4{,}0 = -443{,}64 \ \ kNm$$

Distance of zero shear force location to G: $x_0 = V_{GH} / q$

$$x_0 = 215{,}455 / 40 = 5{,}39 \ \ m$$

Maximum value of moment: $M_{enb.} = M_{GH} + V_{GH}^2 / (2q)$

$$M_{enb.} = 215{,}455^2 / (2 \times 40) = 580{,}26 \ \ kNm$$



$$\cos\alpha = 0{,}447$$
$$\sin\alpha = 0{,}894$$

,

HI member

$$N_{HI} = N_{IH} = 0 \qquad V_{HI} = 40 \times 5{,}0 = 200{,}00 \ \text{kN} \qquad V_{IH} = 0$$

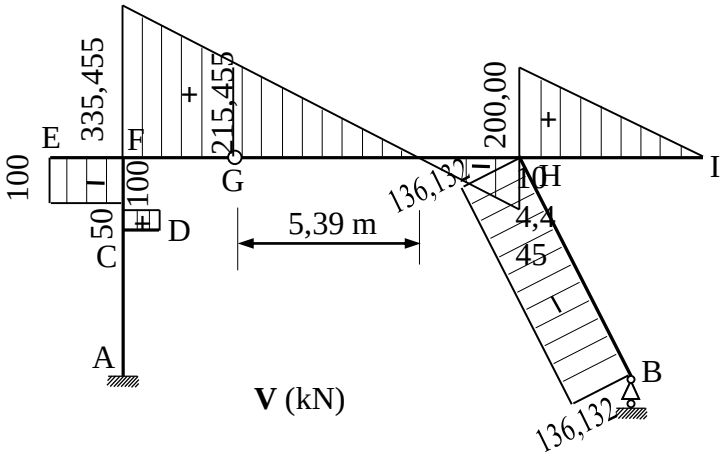
$$M_{HI} = -40 \times 5{,}0^2 / 2 + 30 = -470{,}00 \ \text{kNm} \qquad M_{IH} = 30 \ \text{kNm}$$

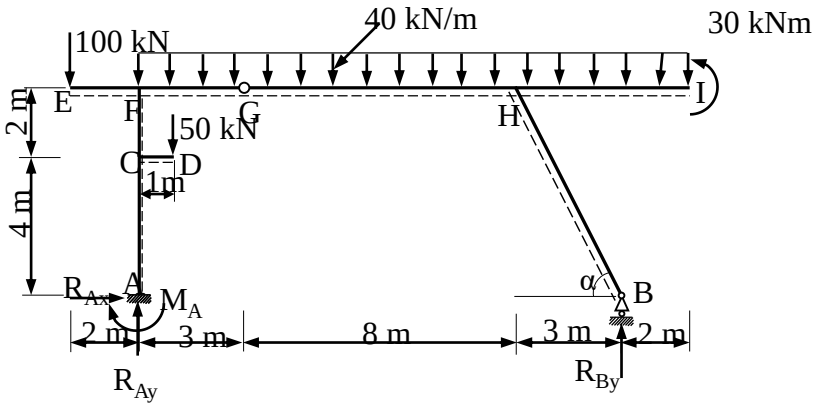
HB member

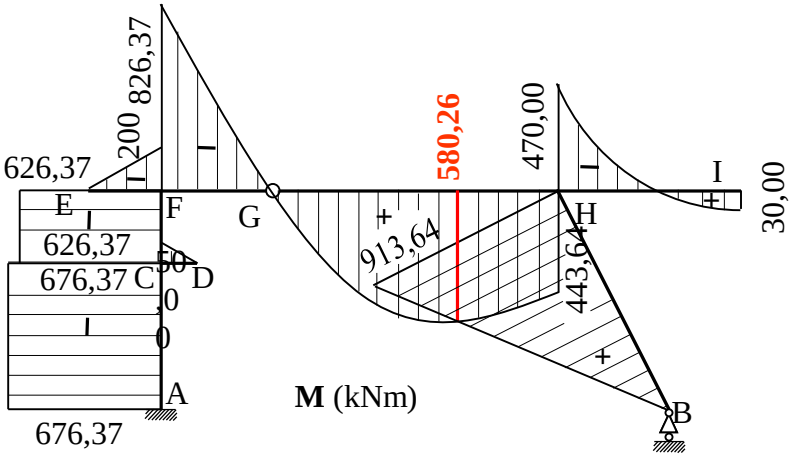
$$N_{HB} = N_{BH} = -304{,}545 \times \sin\alpha = -304{,}545 \times 0{,}894 = -272{,}263 \ \text{kN}$$

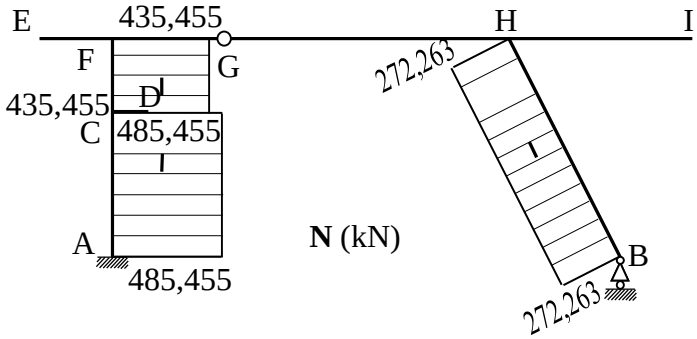
$$V_{HB} = V_{BH} = -304{,}545 \times \cos\alpha = -304{,}545 \times 0{,}447 = -136{,}132 \ \text{kN}$$

$$M_{HB} = 304{,}545 \times 3{,}0 = 913{,}64 \ \text{kNm} \qquad M_{BH} = 0$$

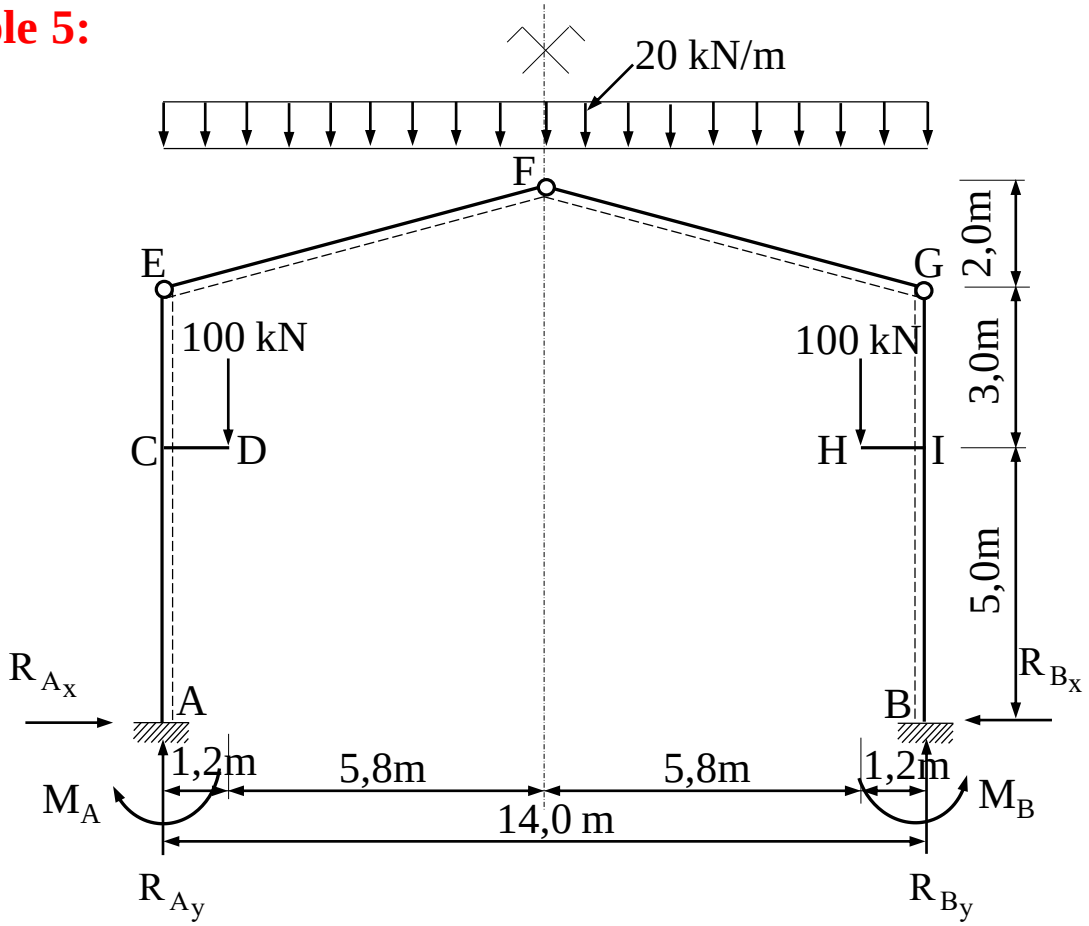

335,455
215,455
200,00
E
F
G
H
I
100
100
50
C
D
5,39 m
136,132
4,4
45
A
B
V (kN)
136,132


40 kN/m
30 kNm
100 kN
2 m
E
F
G
H
I
50 kN
C
D
1m
4 m
R_{Ax}
A
M_A
α
B
2 m
3 m
8 m
3 m
2 m
R_{Ay}
R_{By}

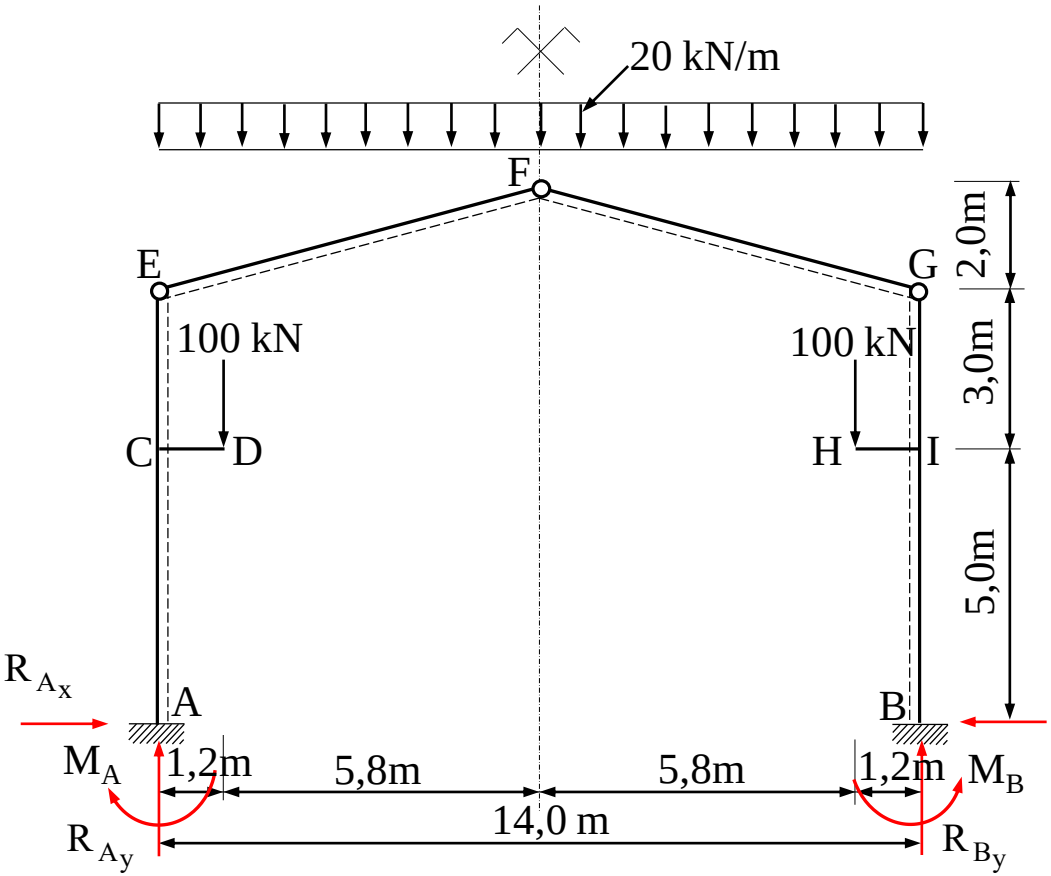

826,37
200
626,37
580,26
470,00
30,00
E
F
G
H
I
626,37
913,64
443,64
676,37
C
50
D
0
0
A
B
M (kNm)
676,37


E
435,455
H
I
F
G
272,263
435,455
D
C
485,455
A
B
N (kN)
485,455
272,263

Example 5:



Estimate support reactions and draw internal force diagrams of the given system.



Solution:

Number of support reactions:

$$n_r = 3 + 3 = 6$$

Number of hinge conditions:

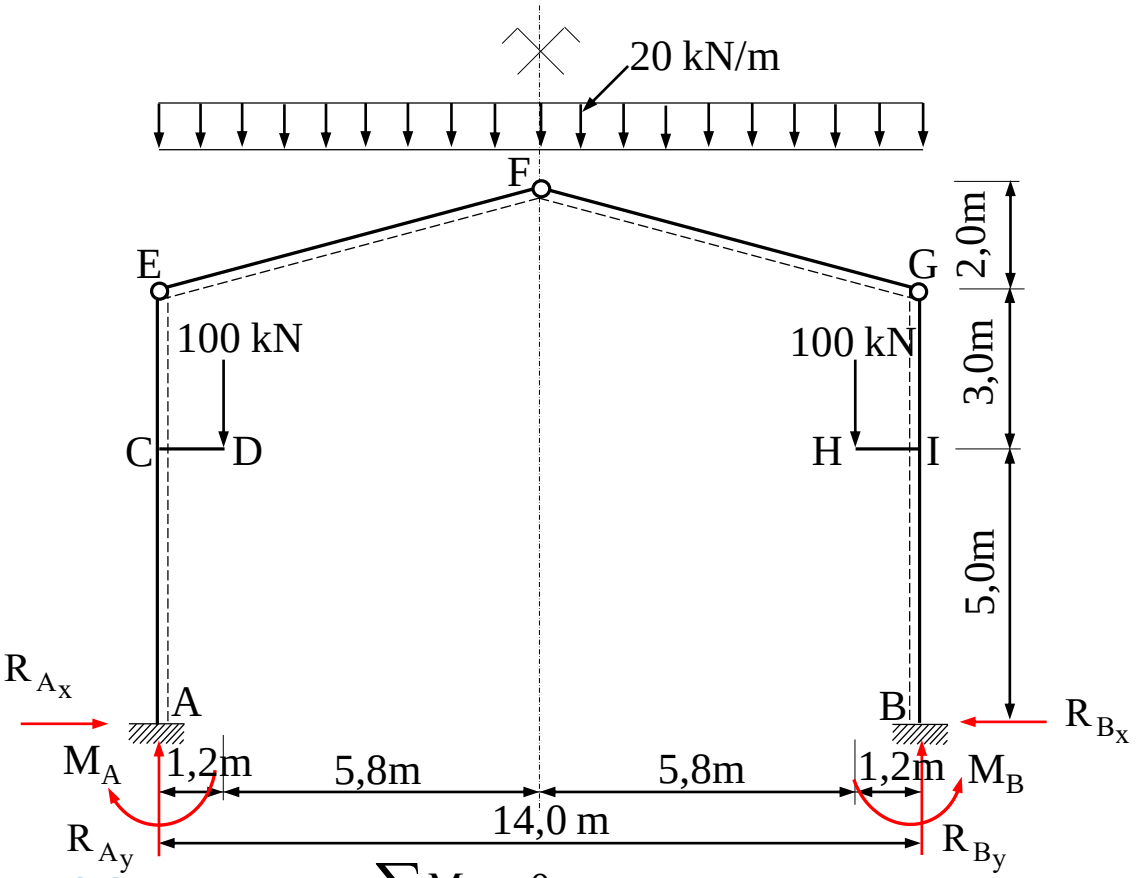
$$n_m = 1 + 1 + 1 = 3$$

$$n = 6 - (3 + 3) = 0$$

Support reactions:

Since the system is symmtrical

$$R_{A_x} = R_{B_x}$$
$$R_{A_y} = R_{B_y}$$
$$M_A = M_B$$



ACE system: $\sum M_E = 0$

$$-R_{Ax} \cdot 8{,}0 + M_A + 100 \times 1{,}2 = 0 \longrightarrow -8R_{Ax} + M_A = -120 \longrightarrow$$

ACEF system $\sum M_F = 0$

$$-R_{Ax} \cdot 10{,}0 + R_{A_y} \cdot 7{,}0 + M_A - 100 \times 5{,}8 - 20 \times 7{,}0 \times 3{,}5 = 0 \searrow$$

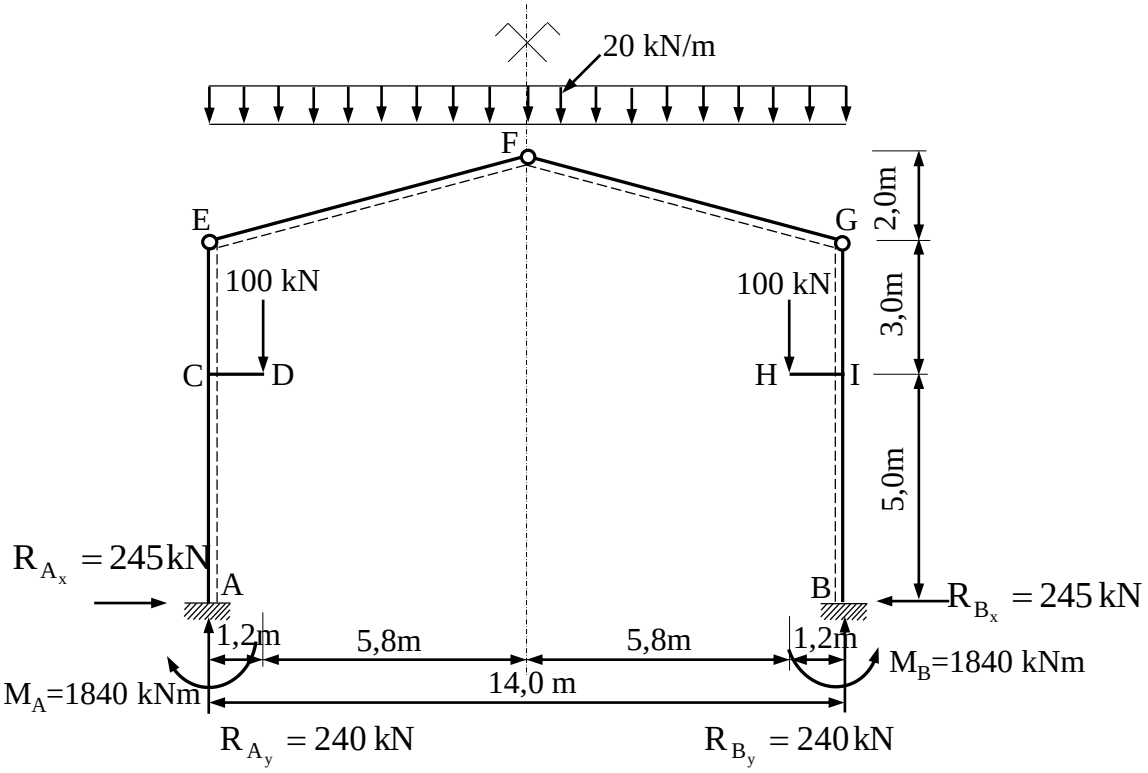
$$-10R_{Ax} + 7R_{A_y} + M_A = 1070 \longrightarrow$$

ACEFG system $\sum M_G = 0$

$$-R_{Ax} \cdot 8{,}0 + R_{A_y} \cdot 14{,}0 + M_A - 100 \times 12{,}8 - 20 \times 14{,}0 \times 7 = 0 \searrow$$

$$-8R_{Ax} + 14R_{A_y} + M_A = 3240 \longrightarrow$$

$$R_{A_x} = 245 \text{ kN} \quad R_{A_y} = 240 \text{ kN} \quad M_A = 1840 \text{ kNm}$$



$$R_{B_x} = R_{A_x} = 245\ \ kN \qquad R_{B_y} = R_{A_y} = 240\ \ kN$$

$$M_B = M_A = 1840\ \ kNm$$

Check the results:

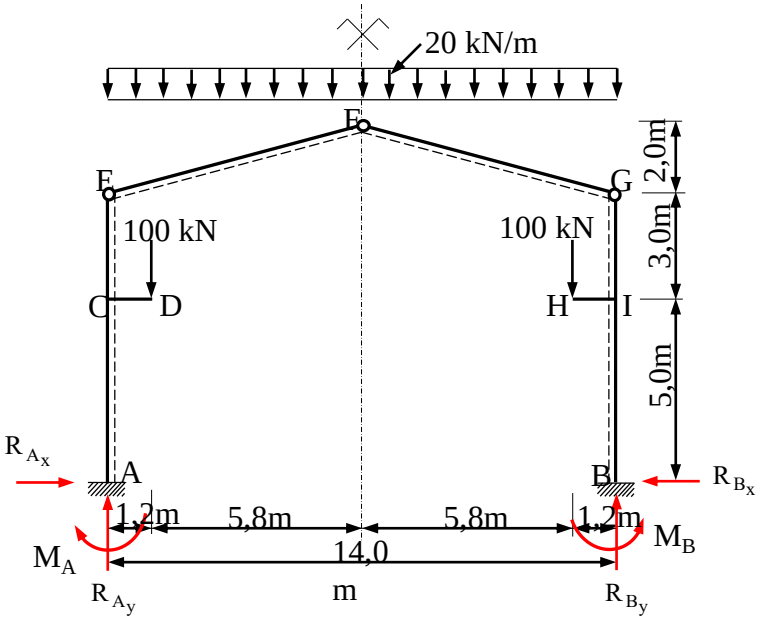
For whole system: $\sum P_y = 0$

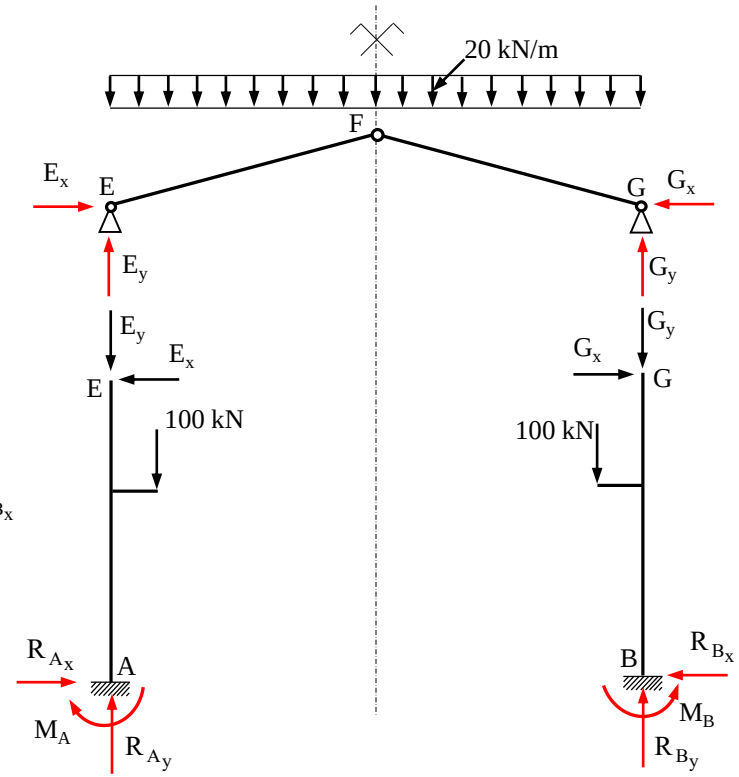
$$\sum P_y = 240 + 240 - 100 - 100 - 20 \times 14 = 0$$

For whole system: $\sum P_x = 0$ $\qquad \sum P_x = 245 - 245 = 0$

For whole system: $\sum M_B = 0$

$$240 \times 14{,}0 + 1840 - 100 \times 12{,}8 - 280 \times 7{,}0 - 100 \times 1{,}2 - 1840 = 0$$





For EFG subsystem: $\sum M_G = 0$

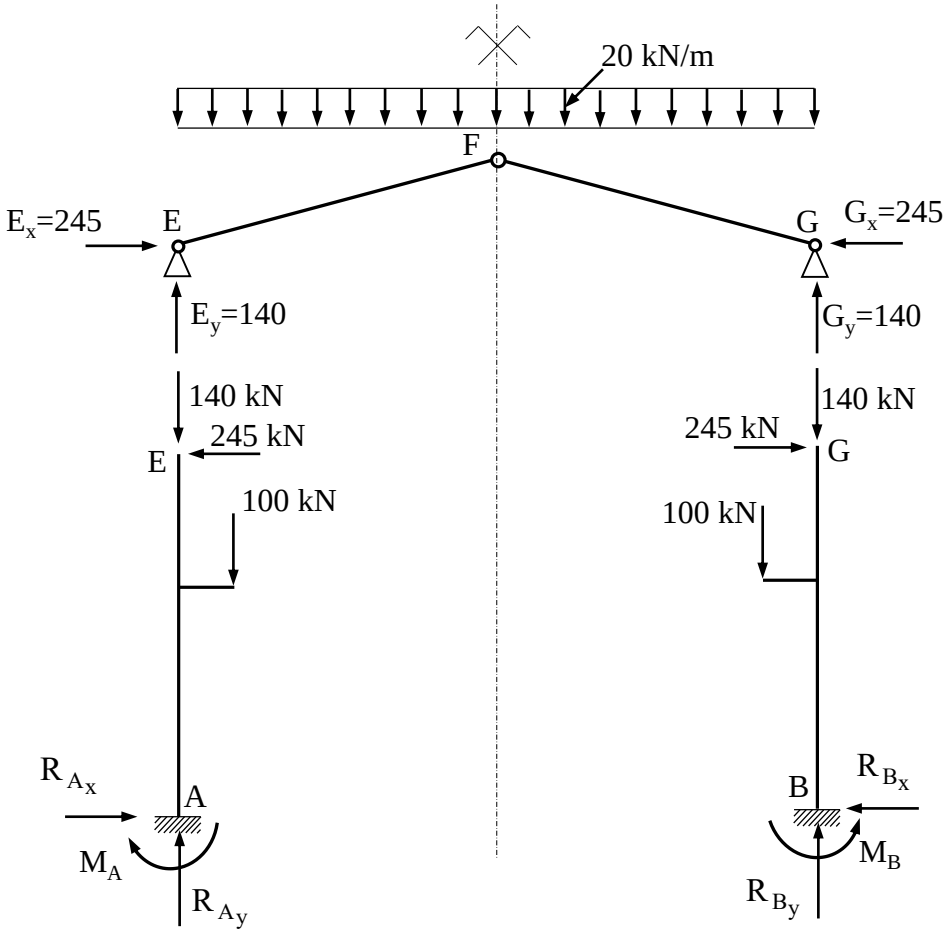
$$E_y \cdot 14{,}0 - 20 \times 14{,}0 \times 7{,}0 = 0 \longrightarrow E_y = 1960/14 = 140 \text{ kN}$$

F section of EF member of EFG subsystem $\sum M_F = 0$

$$E_y \cdot 7{,}0 - E_x \cdot 2{,}00 - 20 \times 7{,}0 \times 3{,}5 = 0$$

$$7E_y - 2E_x = 490 \longrightarrow E_x = (140 \times 7 - 490)/2 = 245 \text{ kN}$$

Using symmetry rule $G_x = E_x = 245$ kN $\quad G_y = E_y = 140$ kN



Reactions of support E of EFG system are applied to the E point of ACE system:

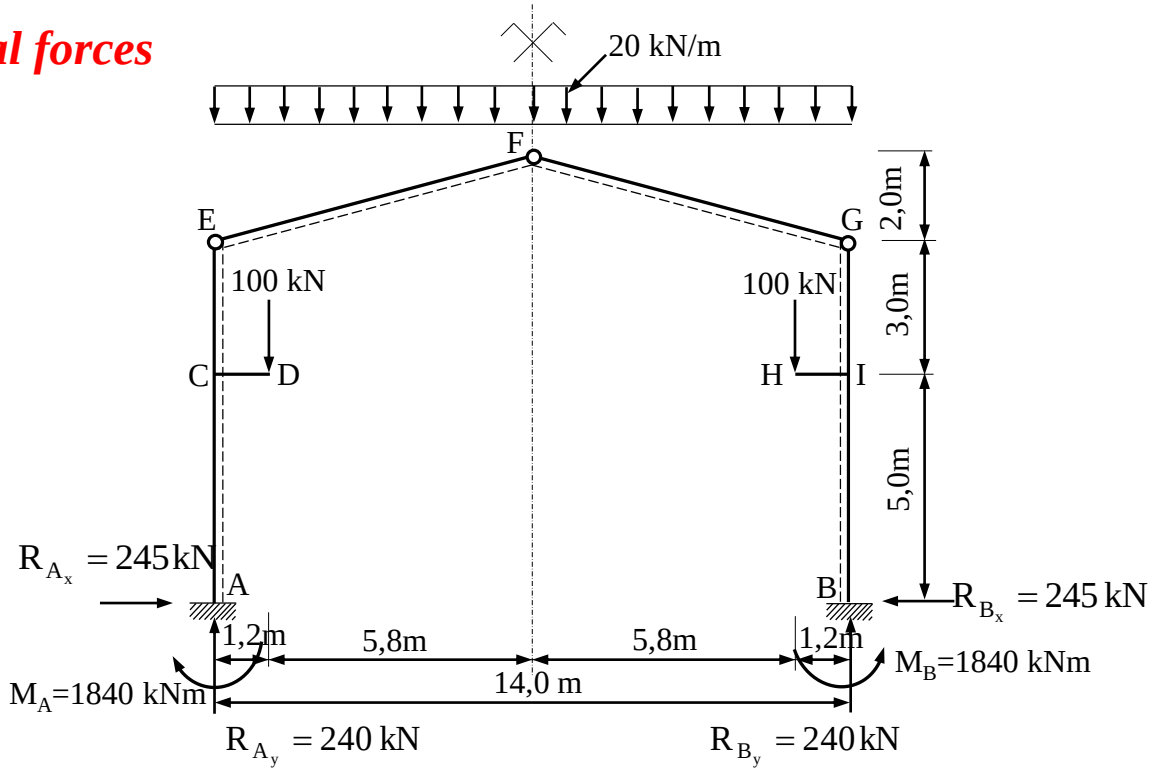
$$\sum P_x = 0 \longrightarrow R_{A_x} = 245 \ \text{kN} \qquad \sum P_y = 0 \longrightarrow R_{A_y} = 140 + 100 = 240 \ \text{kN}$$

$$\sum M_A = 0 \longrightarrow M_A = 245 \times 8{,}0 - 100 \times 1{,}2 = 1840 \ \text{kNm}$$

Obtaining the reactions of support B using symmetry condition

$$R_{B_x} = R_{A_x} = 245 \ \text{kN} \qquad R_{B_y} = R_{A_y} = 240 \ \text{kN} \qquad M_B = M_A = 1840 \ \text{kNm}$$

Internal forces



$$L_{EF} = L_{FG} = \sqrt{7^2 + 2^2} = 7{,}28 \ \ m$$

$$\sin\alpha = 2/7{,}28 = 0{,}2747$$

$$\cos\alpha = 7/7{,}28 = 0{,}9615$$

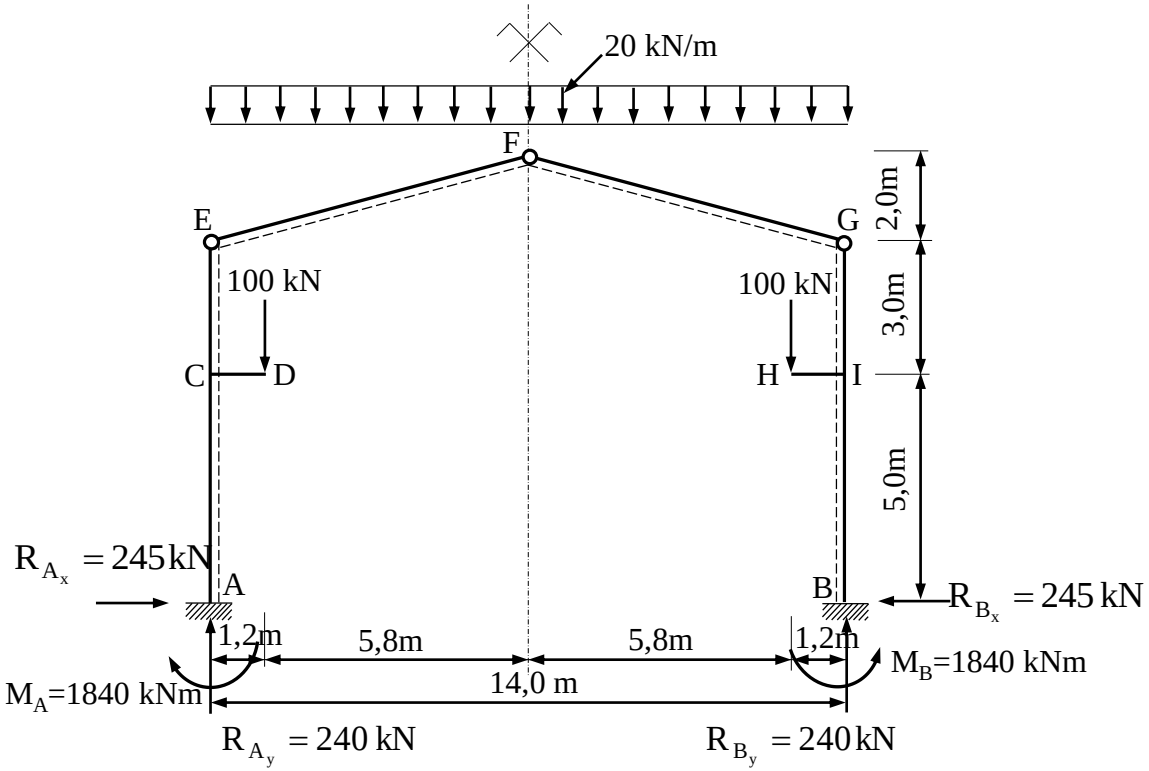
Since the system is symmetrical, internal forces of ACEF system are determined and BIGF system will have the same magnitude and sign of moment and axial force with the ACEF system, but BIGF system will have same magnitude and opposite sign of shear force with ACEF system.

AC member

$$N_{AC} = N_{CA} = -240{,}000 \ \ kN \qquad V_{AC} = V_{CA} = -245{,}000 \ \ kN$$

$$M_{AC} = 1840{,}00 \ \ kNm$$

$$M_{CA} = 1840{,}00 - 245{,}00 \times 5{,}0 = 615{,}00 \ \ kNm$$



CD member

$$N_{CD} = N_{DC} = 0 \qquad V_{CD} = V_{DC} = 50{,}000 \ \ kN$$

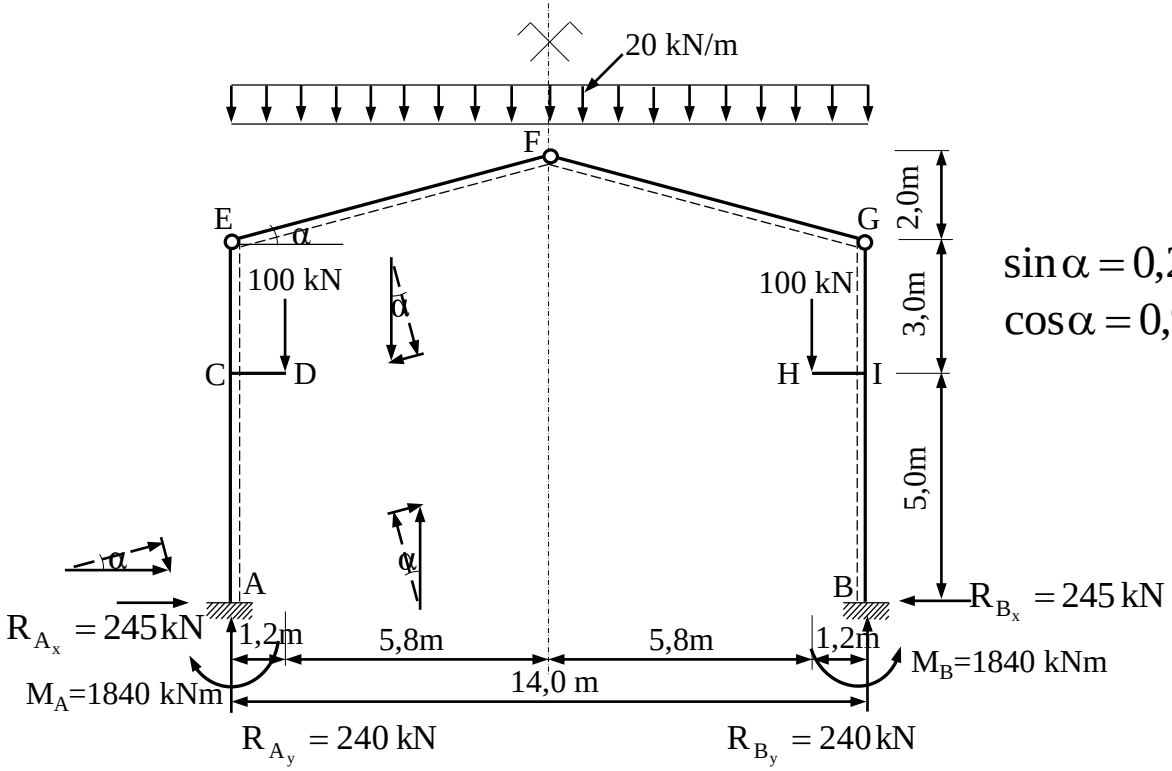
$$M_{CD} = -100 \times 1{,}20 = -120{,}00 \ \ kNm \qquad M_{CD} = 0$$

CE member

$$N_{CE} = N_{EC} = -240 + 100 = -140{,}000 \ \ kN$$

$$V_{CE} = V_{EC} = -245{,}000 \ \ kN \qquad M_{EC} = 0$$

$$M_{CE} = 1840 - 245 \times 6{,}0 + 100 \times 1{,}2 = 735{,}00 \ \ kNm$$



$\sin\alpha = 0{,}2747$

$\cos\alpha = 0{,}9615$

EF member

$$N_{EF} = -(240-100)\cdot\sin\alpha - 245\cdot\cos\alpha$$

$$N_{EF} = -140\times 0{,}2747 - 245\times 0{,}9615 = -274{,}026\ \ kN$$

$$V_{EF} = (240-100)\cdot\cos\alpha - 245\cdot\sin\alpha$$

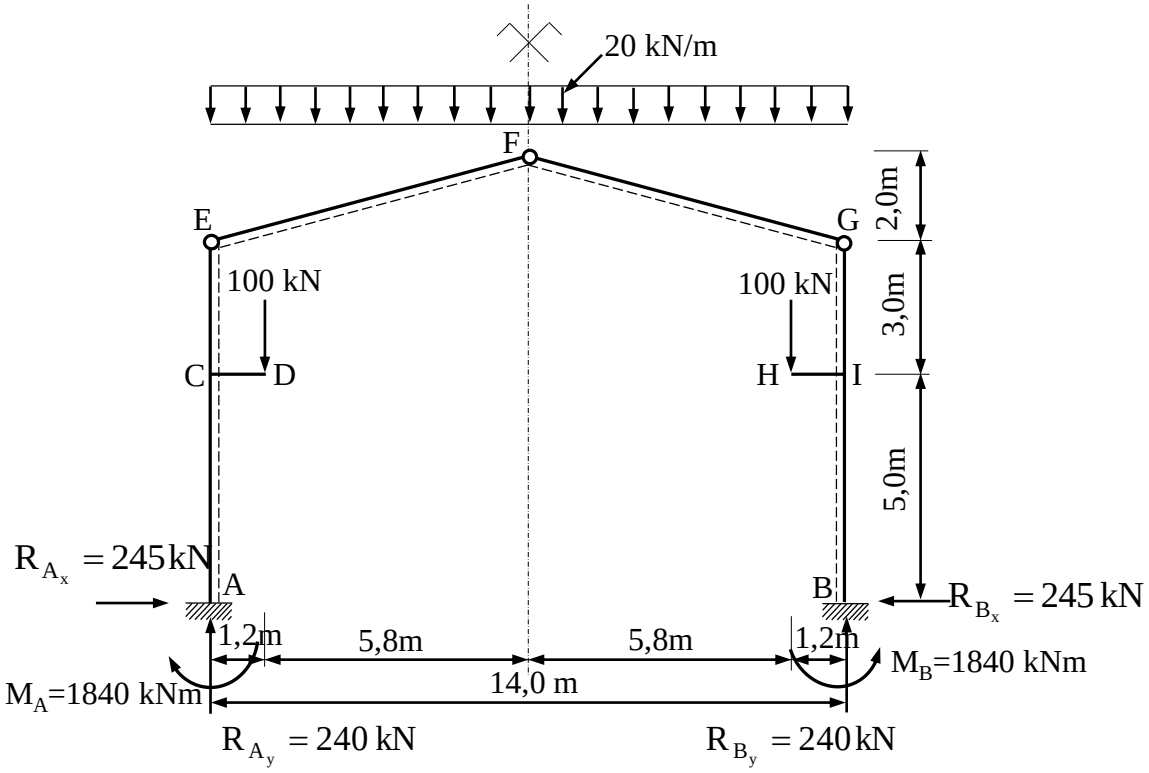
$$V_{EF} = 140\times 0{,}9615 - 245\times 0{,}2747 = 67{,}302\ \ kN \qquad M_{EF} = 0$$

$$N_{FE} = -(240-100-20\times 7{,}0)\cdot\sin\alpha - 245\cdot\cos\alpha$$

$$N_{FE} = -245\times 0{,}9615 = -235{,}57\ \ kN$$

$$V_{FE} = (240-100-20\times 7{,}0)\cdot\cos\alpha - 245\cdot\sin\alpha$$

$$V_{FE} = -245\times 0{,}2747 = -67{,}302\ \ kN \qquad M_{FE} = 0$$



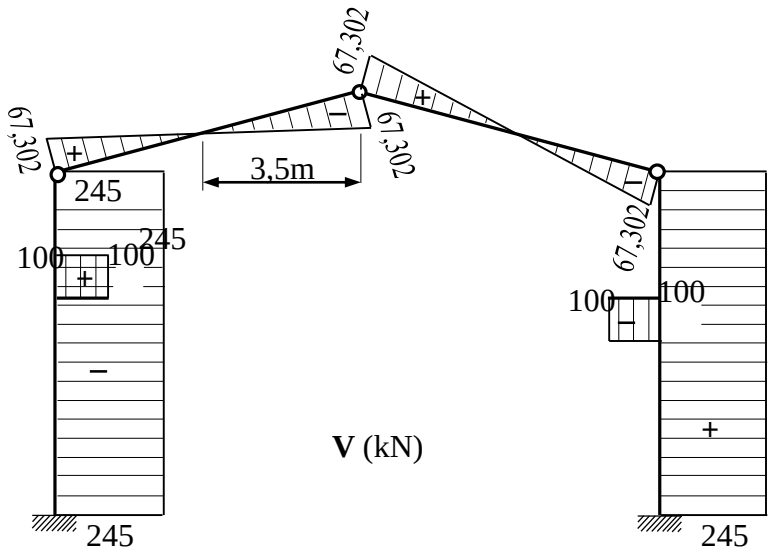
The distance of zero shear force location to point E:

$$V_x = 67{,}392 - 20x \cdot x / \cos\alpha = 0$$

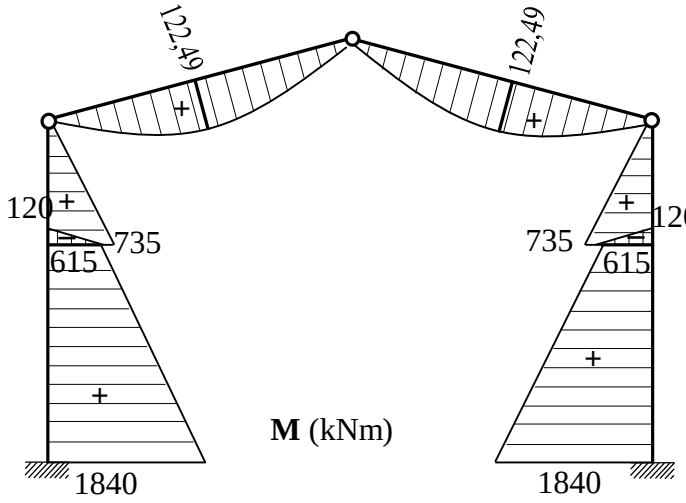
$$x_0 = 67{,}302/(20 \times 0{,}9615) = 3{,}50 \ \ m$$

Maximum moment on EF member

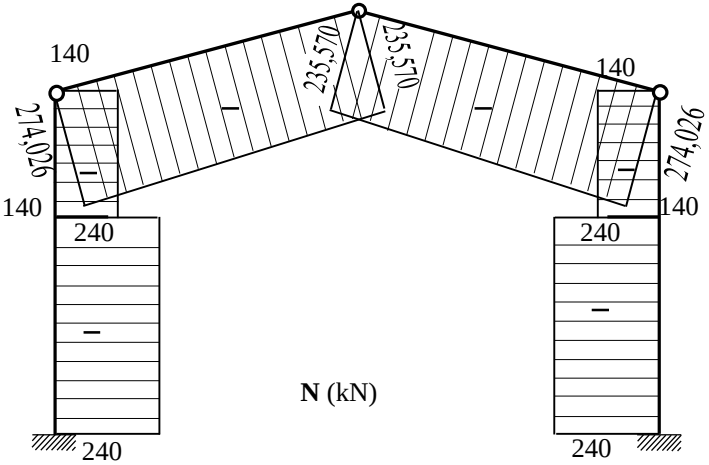
$$M_{enb.} = 67{,}302 \times (7{,}28/2) - 20 \times 3{,}50 \times 1{,}75 = 122{,}49 \ \ kNm$$


67,302
67,302
67,302
67,302
3,5m
245
245
100
100
100
100
245
245

V (kN)

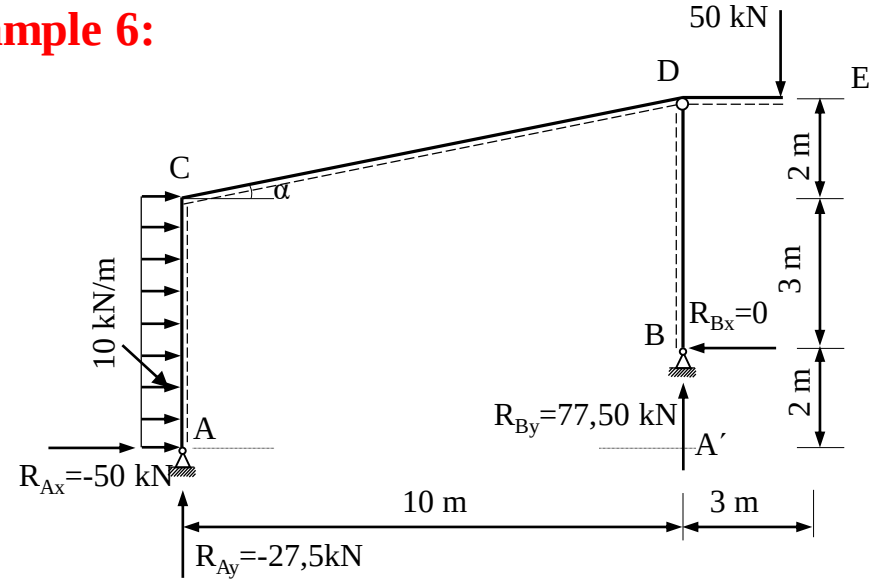

122,49
122,49
120
735
615
735
615
120
1840
1840

M (kNm)


140
140
235,570
235,570
274,026
274,026
140
140
240
240
240
240

N (kN)

Example 6:



Estimate support reactions and draw internal force diagrams of the given system.

Solution:

$$L_{CD} = \sqrt{100+4} = 10{,}198 \text{ m}$$

$$\sin\alpha = 2/10{,}198 = 0{,}196$$

$$\cos\alpha = 10/10{,}198 = 0{,}981$$

Support reactions

$\sum M_D = 0$ at point D of BD frame due to hinge condition

$$R_{BX} \cdot 5 = 0 \longrightarrow R_{BX} = 0$$

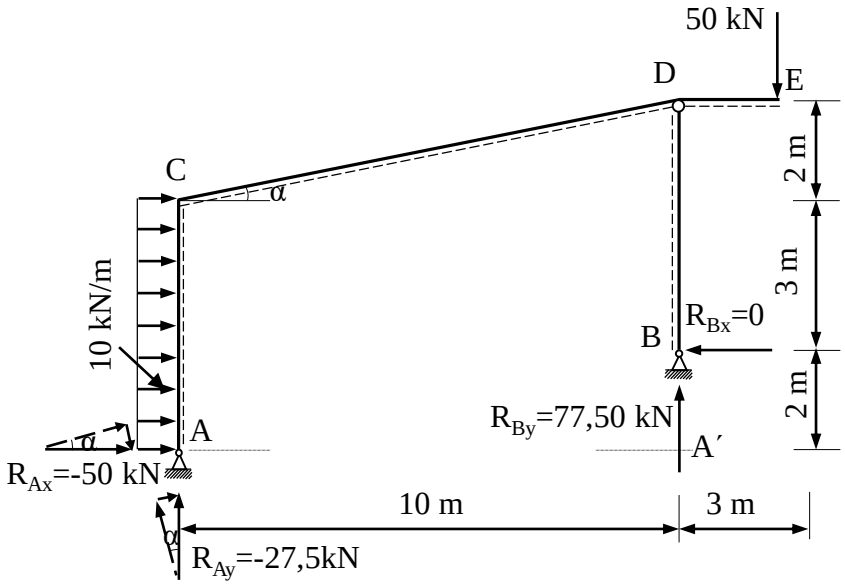
For the system $\sum P_x = 0 \longrightarrow R_{Ax} = -10 \times 5{,}0 = -50 \text{ kN}$

Moment respect to point A´ where R_{Ax} and R_{By} are intersected

$$\sum M_{A'} = 0 \longrightarrow R_{Ay} = (-10 \times 5{,}0 \times 2{,}5 - 50 \times 3{,}0)/10 = -27{,}50 \text{ kN}$$

For the system $\sum M_A = 0$

$$R_{By} = (10 \times 5{,}0 \times 2{,}5 + 50 \times 13{,}0)/10 = 77{,}50 \text{ kN}$$



Check the results: $\sum P_y = -27{,}5 + 77{,}5 - 50 = 0$

Internal forces

AC frame

$N_{AC} = N_{CA} = 27{,}50 \ \ kN \qquad V_{AC} = -(-50) = 50 \ \ kN \qquad M_{AC} = 0$

$V_{CA} = -(-50) - 10 \times 5{,}0 = 0$

$M_{CA} = -(-50) \times 5{,}0 - 10 \times 5{,}0 \times 2{,}5 = 125 \ \ kNm$

CD frame

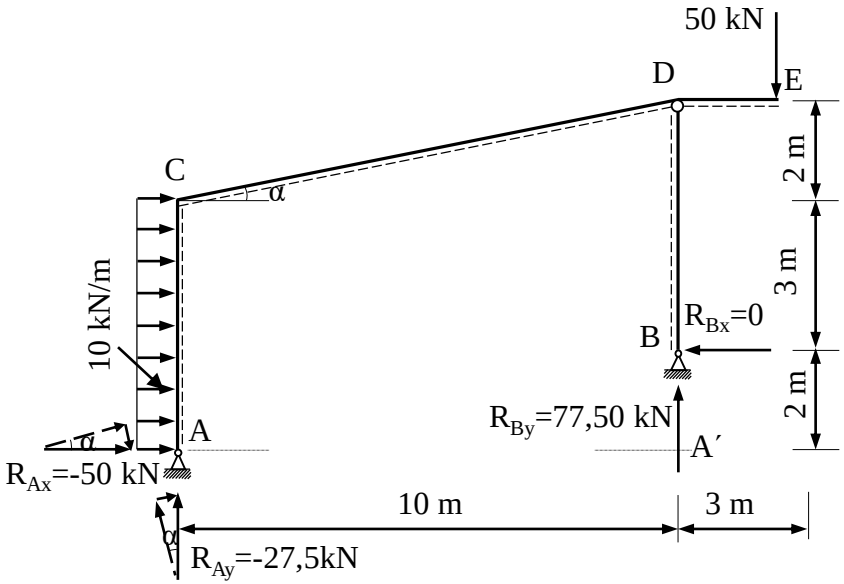
$N_{CD} = N_{DC} = -(-50)\cos\alpha - (-27{,}50)\sin\alpha - 10 \times 5{,}0 \cdot \cos\alpha$

$N_{CD} = N_{DC} = 27{,}50 \times 0{,}196 = 5{,}39 \ \ kN$

$V_{CD} = V_{DC} = -(-50)\sin\alpha + (-27{,}50)\cos\alpha - 10 \times 5{,}0 \cdot \sin\alpha$

$V_{CD} = V_{DC} = -27{,}50 \times 0{,}981 = -26{,}98 \ \ kN$

$M_{CD} = M_{CA} = 125 \ \ kNm$

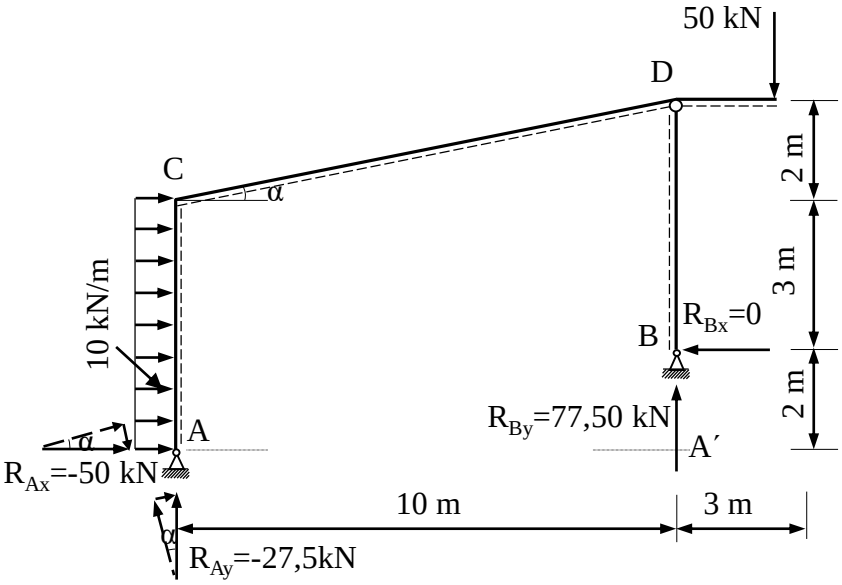


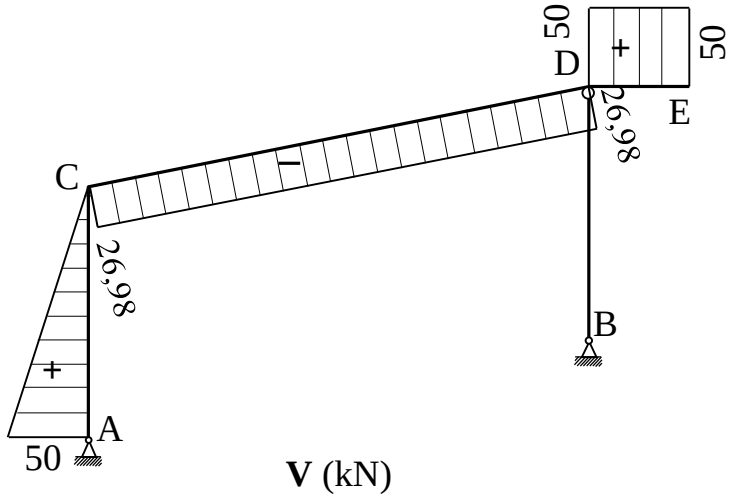
DE frame

$$N_{DE} = N_{ED} = 0 \qquad V_{DE} = V_{ED} = 50 \ \text{kN} \qquad M_{ED} = 0$$

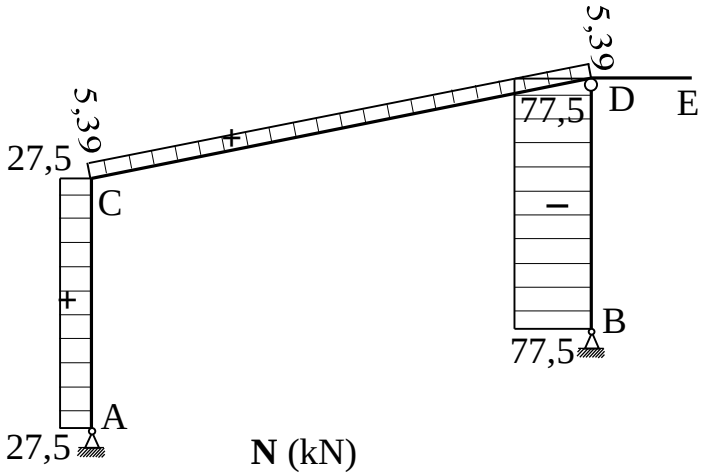
BD frame

$$N_{BD} = N_{DB} = -77{,}50 \ \text{kN} \qquad V_{BD} = V_{DB} = 0 \qquad M_{BD} = M_{DB} = 0$$

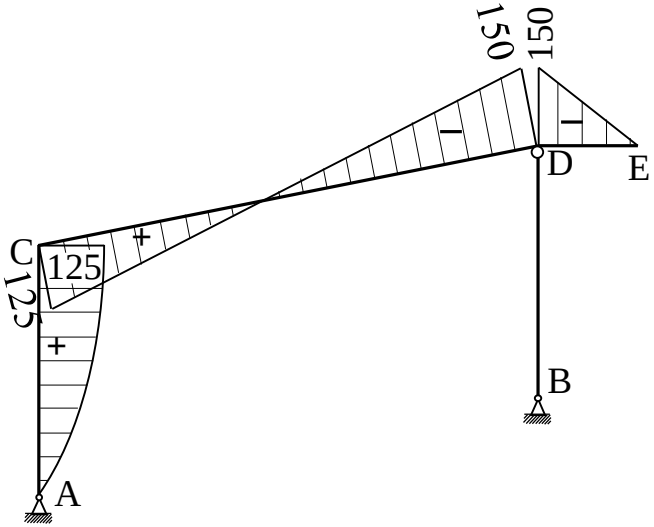

50 kN
D
C
α
2 m
3 m
2 m
10 kN/m
B
R_{Bx}=0
R_{By}=77,50 kN
A
A´
α
R_{Ax}=-50 kN
10 m
3 m
α
R_{Ay}=-27,5kN


50
50
D
+
E
26,98
C
26,98
B
+
A
50

V (kN)


5,39
5,39
D
E
77,5
27,5
C
+
−
B
77,5
+
A
27,5

N (kN)


150
150
D
E
C
125
125
+
B
+
A

M (kNm)

Reference:

- Prof. İbrahim EKİZ, YAPI STATİĞİ 1, BİRSEN YAYINEVİ

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